FOREWORD

This is the second iteration of preparing our own courseware material, after successful completion of a similar task undertaken a few years ago. These contents have been carefully prepared and should serve as excellent auxiliary material for both instructors and students.

The Special Academic Group, Autonomy (SAGA) was formed for the sole purpose of preparing courseware contents primarily in the first-year theory subjects; a few second-year subjects were also included. The subjects for the first year were - Basic Electrical Engineering, Basic Electronics Engineering, Computer Programming, Data Structures and Algorithms, Engineering Mathematics I and II, Engineering Physics, Engineering Chemistry, Constitution of India, Environmental Science and Engineering and Communicative and Technical English. For the second year, the subjects for which courseware material was prepared were Analog Electronic Circuits, Digital Systems Design, Circuit Theory and Measurements and Instruments.
Faculty members from all the departments contributed to the task. They were, in no particular order, Nalini Singh, Bimal Meher, Saumyaranjan Dash, Mukti Routray, Susmita Biswal, Manasa Dash, Bipin Tripathy, Sibasankar Nayak, Janmejay Senapati, Subrat Sahu, Pradeep Moharana, Rupambika Pattanaik, Dhananjay Tripathy, Jagadish Patra, Sachin Das, Deepak Ranjan Nayak, Amulya Roul, Bodhisattva Dash, Sanghamitra Das, Gyana Ranjan Biswal, Nibedita Swain and Rajan Mishra.

The entire group worked diligently to successfully complete the task which included a peer review of the material. I take this opportunity to thank all the members of the SAGA group for a job well done.

I sincerely hope that this courseware material comes in handy and is utilized to the fullest extent. These are readily available additional resources prepared in accordance with the Silicon autonomy syllabus, to complement textbooks and classroom lectures. If there are any errors, I would be grateful if they are brought to my notice so that we can correct them in subsequent versions.

Dr. Jaideep Talukdar, Principal
Silicon Institute of Technology
Bhubaneswar
December 2020

Module – 1: WATER TREATMENT

Hardness:

Hardness of water is a characteristic property by which water *"'prevents lathering of soap".*

Causes:

- The hardness of water is due to presence of certain salts (mainly bicarbonates, sulphates and Chlorides) of Ca, Mg and few other heavy metal salts dissolved in water. A sample of hard water when treated with soap (sodium or potassium salt of higher fatty acid like oleic, palmitic or stearic) do not produce lather with soap, but on the other hand forms a white scum or precipitate.

- The precipitate is formed due to insoluble soaps of Ca and Mg.

Reactions:-

$$2C_{17}H_{35}COONa + CaCl2 \rightarrow Ca(C_{17}H_{35}COO)_2 + NaCl$$

Sodium Stearate (Sodium soap) hardness Calcium Stearate (Insoluble)

$$2\ C_{17}H_{35}COONa + MgSO4 \rightarrow (C_{17}H_{35}COO)_2Mg + Na_2SO_4$$

Magnesium Stearate (Insoluble)

Thus, the water which does not produce lather with soap solution readily is called Hard water the one which lathers easily on shaking with soap solution is called **soft water.**

<u>Causes of Hardness:</u>

Hardness of water is due to the presence of Bicarbonates, Chlorides, Sulphates and Nitrates of Calcium and Magnesium. These soluble salts get mixed with natural water due to the following reasons:

1. When natural water containing CO2 flows over the rocks of the limestone ($CaCO_3$) and Dolomite ($CaCO_3$ & $MgCO_3$), they get converted into soluble bicarbonates. Thus, water gets hardness.

$$CaCO_3 + H2O + CO2 \rightarrow Ca\ (HCO3)2$$

Insoluble Soluble

2. When natural water flows over the rocks containing chlorides and sulphates and Nitrates of Calcium and magnesium, these salts dissolve in water. Thus water gets hardness.

Disadvantages of Hard Water:

1. **In Domestic use:**

> Washing: Hard water, when used for washing purposes, does not producing lather freely with soap. As a result cleaning quality of soap is decreased and a lot of it is wasted.

> Bathing: Hard water does not lather freely with soap solution, but produces sticky scum on the bathtub and body. Thus, the cleaning quality of soap is depressed and a lot of it is wasted.

> Cooking: The boiling point of water is increased because of presence of salts. Hence more fuel and time are required for cooking.

> Drinking: Hard water causes bad effects on our digestive system. Moreover, the possibility of forming calcium oxalate crystals in urinary tracks is increased.

2. **Industrial Use:**

> Textile Industry: Hard water causes wastage of soap. Precipitates of calcium and magnesium soaps adhere to the fabrics and cause problem.

> Sugar Industry: The water which containing sulphates, nitrates, alkali carbonates are used in sugar refining, cause difficulties in the crystallization of sugar.

> Dyeing Industry: The dissolved salts in hard water may reacts with costly dyes forming precipitates.

> Paper Industry: Calcium, magnesium, Iron salts in water may affect the quality of paper.

> Pharmaceutical Industry: Hard water may cause some undesirable products while preparation of pharmaceutical products.

3. **Steam generation in Boilers:**

> For steam generation, boilers are almost invariably employed. If the hard water is fed directly to the boilers, there arise many troubles such as: Scales & sludges formation, Corrosion, Priming & Foaming and Caustic embrittlement.

TYPES OF HARDNESS

Hardness of water is mainly two types:
1. Temporary Hardness 2. Permanent Hardness

1. Temporary Hardness: Temporary Hardness mainly caused by the presence of dissolved

bicarbonates of Calcium, Magnesium (Ca (HCO3)2, Mg (HCO3)2). Temporary Hardness can be largely removed by boiling of water.

$$Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2$$

Calcium bicarbonate

$$Mg(HCO_3)_2 \rightarrow Mg(OH)_2 + 2CO_2$$

Magnesium bicarbonate

2. Permanent Hardness: It is due to the presence of dissolved Chlorides, Nitrates and Sulphates of Calcium, Magnesium, Iron and other metals. Permanent hardness responsible salts are $CaCl_2$,, $MgCl_2$, CaSO4, MgSO4, FeSO4, $Al_2(SO_4)_3$. Permanent Hardness cannot be removed by boiling but it can be removed by the use of *chemical agents.*

EXPRESSION AND UNITS OF HARDNESS

The expression of hardness producing salts usually expressed in terms of an equivalent amount of CaCO3. Calcium Carbonate is chosen as a standard because:

i. Its molecular weight (100) and equivalent weight (50) is a whole number, so the calculations in water analysis can be simplified.
ii. It is the most insoluble salt that can be precipitated in water treatment.

The conversion of the hardness causing salts into $CaCO_3$ equivalents can be achieved by using the following formula:

$$\text{Degree of Hardness} = \text{The weight of hardness causing salts} \times \frac{100\ (\text{Molecular weight of } CaCO_3)}{}$$

Units of Hardness:

1. Parts per Million (ppm): The number of parts of calcium carbonate equivalent hardness presents in 10^6 parts of water.

1ppm = 1 part of $CaCO_3$ eq hardness in 10^6 parts of water.

2. Milligrams per litre (mg/1): The number of milligrams of calcium carbonate equivalent hardness presents in litre of water.

1 mg/L = 1 mg of $CaCO_3$ eq hardness in 1 litre of water.

But one litre of water weights =1 kg =1000g = 1000 x 1000 mg = 10^6 mg = 1 ppm.

3. Clark's degree (°Cl): The number of parts of calcium carbonate equivalent hardness presents in 70,000 or (7×10^4) parts of water.

1° Clarke = 1 part of $CaCO_3$ eq hardness per 70,000 parts of water.

4. Degree French (°Fr): The number of parts of calcium carbonate equivalent hardness presents in 10^5 parts of water.

1° Fr = 1 part of $CaCO_3$ hardness eq per 10^5 parts of water.

5. Milli-equivalent per liter (meq/L):- is the number of milli-equivalents of hardness present per liter.

1meq/L= 1meq of $CaCO_3$ per liter of water = 10^{-3}x 50 g of $CaCO_3$ eq. per liter

= 50 mg of $CaCO_3$ eq. per liter = 50 mg/L of $CaCO_3$ eq. = 50ppm.

Relationship between various units of hardness:

1 ppm	= 1 mg/L	= 0.1° Fr	= 0.07° Cl = 0.02 meq/ L
1 mg/L	= 1 ppm	= 0.1° Fr	= 0.07° Cl
1 ° Cl	= 1.433° Fr	= 14.3 ppm	= 14.3 mg/L
1 ° Fr	= 10 ppm	= 10 mg/L	= 0.7° Cl

calculation of equivalents of $CaCO_3$:

Salt/ion	Molar mass	Chemical Equivalent or Equivalent Weight	Multiplication factor for converting into equivalents of $CaCO_3$
$Ca(HCO3)_2$	162	81	100/162
$Mg(HCO3)_2$	146	73	100/146
$CaSO_4$	136	68	100/136
$CaCl_2$	111	55.5	100/111
$MgSO_4$	120	60	100/120
$MgCl_2$	95	47.5	100/95
$CaCO_3$	100	50	100/100
$MgCO_3$	84	42	100/84
CO_2	44	22	100/44
$Ca(NO_3)_2$	164	82	100/164
$Mg(NO_3)_2$	148	74	100/148
HCO_3^-	61	61	100/122
OH^-	17	17	100/34
CO^2	60	30	100/60
$NaAlO_2$	82	82	100/164
$Al_2(SO_4)_3$	342	57	100/114
$FeSO_4.7H_2O$	278	139	100/278
H^+	1	1	100/2
HCl	36.5	1	100/73

Problem-1: A sample of water is found to contains following dissolving salts in milligrams per litre $Mg(HCO_3)_2$ = 73, $CaCl_2$ = 111, $Ca(HCO_3)_2$ = 81, $MgSO_4$ = 40 and $MgCl_2$ = 95. Calculate temporary and permanent hardness and total hardness.

Solution:

Name of the hardness causing salts	Amount of the hardness causing salts(mg/Lit)	Molecular weight of hardness causing salts	Amounts equivalent to CaCO3 (mg/Lit)
$Mg(HCO_3)_2$	73	146	73x100/146 = 50
$CaCl_2$	111	111	111x100/111 = 100
$Ca(HCO_3)_2$	81	162	81x100/162 = 50
$MgSO_4$	40	120	40x100/120 = 33.3
$MgCl_2$	95	95	95x100/95 = 100

Temporary hardness = $Mg(HCO3)_2 + Ca(HCO3)_2$

= 50 + 50 = 100mgs/Lit.

Permanent hardness = $CaCl_2 + MgSO_4 + MgCl_2$

= 100 + 33.3 + 100 = 233.3mgs/Lit.

Total hardness = Temporary hardness + Permanent hardness = 100 + 233.3 = 333.3mgs/Lit.

POTABLE WATER AND ITS SPECIFICATIONS

Water free from contaminants or water that is safe for human consumption is called potable water. The following are the specifications of water drinking purpose.

1. The water should be clear (colorless), odorless and pleasant taste.
2. The optimum *hardness* of water must be 125ppm.
3. The *pH* of potable water should be 7.0 to 8.5.
4. The recommended maximum concentration of *total dissolved solids (TDS)* in potable water must not exceed 500 ppm.
5. The *turbidity* in drinking water should not exceed 25 ppm.
6. The water must be free from *heavy metals* like Lead, Arsenic, Chromium and Manganese.
7. The water must be *free from pathogenic bacteria*
8. The water must be *free from dissolved gases* like H2S, CO2 and NH3.

BOILER TROUBLES

A boiler is a closed vessel in which water under pressure is transformed into steam by the application of heat. The steam so generated is used in industries and generation of power. In modern pressure boilers and laboratories, the water required is used pure than the distilled water.

A boiler feed water should correspond with the following composition:

> Its hardness should be below 0.2ppm.
> Its caustic alkalinity (due to OH^-) should lie between 0.15ppm to 0.45ppm.
> Its soda alkalinity (due to Na_2CO_3) should lie in between 0.45ppm to 1ppm.
> It's should be free from dissolved gases like O2, CO2, in order to prevent boiler corrosion.

Excess of impurities in the boiler feed water generally cause the following problems:

1. Sludge's and Scale formation 2. Caustic embrittlement

Boilers are employed for the steam generation in power plants, where water as continuously heated to produce steam. As more and more water is removed from water in the form of steam, the boiler water gets concentrated with dissolved salts progressively reaches the saturation point. At this point the dissolved salts are precipitated out and slowly settle on the inner walls of the boiler plate. The precipitation takes place in two ways.

SLUDGES

Sludge is a soft, loosy and slimy precipitate formed within the boiler. It is formed at comparatively colder portions of the boiler and collects in the area where flow rate is slow.

Ex: $MgCO_3$, MgCl2, $CaCl_2$, $MgSO_4$.

Reasons for formation of sludges:

The dissolved salts whose solubility is more in hot water and less in cold water produce sludges.

Disadvantages of sludges:

1. Sludges are bad conductors of heat and results in the wastage of heat and fuel.
2. Excessive sludge formation leads to the settling of sludge in slow circulation areas such as pipe connections, plug openings, gauge-glass connections leading to the choking of the pipes.

Prevention of sludge formation:

a. By using soft water which is free from dissolved salts like $MgCO_3$, $MgCl_2$, CaC_{l2} and MgSO4 can prevent sludge formation.
b. By blow down operation carried out frequently can prevent sludge formation.

Sludges and scales in boiler

SCALES

Scales are hard, adhering precipitates formed on the inner walls of the boilers. Scales are stick very firmly on to the inner walls of the boiler. It is removed with chisel and hammer. Scales are formed by decomposition of calcium bicarbonate in low pressure boilers. It is most due to the presence of CaSO4, $MgCl_2$, Silica, $Ca(HCO)_2$.

Reasons for formation of scales:

a. Decomposition of calcium bicarbonate: The calcium bicarbonate at high temperature decomposes to calcium carbonate which is insoluble salt, forms scale in low pressure boilers.

$Ca(HCO_3)_2 \rightarrow CaCO_3 + H2O + CO2$

b. Hydrolysis of Magnesium salts: Magnesium salts gets hydrolyzed at high temperature forming Mg(OH)2 precipitation which forms salt type scale.

$$MgCl_2 + H2O \rightarrow Mg(OH)_2 + 2HCl$$

c. Decomposition of calcium sulphate: The solubility of CaSO4 in water decreases with the increase in temperature and forms precipitation on the surface of the boiler further which forms hard scale. This type of scales is formed in high-pressure boilers.

d. Presence of silica: SiO2 present even in small quantities, deposits as Calcium silicates (CaSiO3) or Magnesium silicates (MgSiO3). The deposits form hard scale and are very difficult to remove.

Disadvantages of Scales:

Wastage of heat and fuels: Scales poor thermal conductivity so that rate of heat transformation is Thickness (mm)) 0.325 0.625 1.25 2.5 12

Wastage of fuel 10% 15% 50% 80% 150%

1. Lowering of boiler safety is due to overheating of the boiler material becomes softer and weaker, which causes distortion of boiler.
2. Decrease in efficiency of the boiler due to scales deposited in the values and condensers of the boiler cause choking.
3. Danger of explosion which happens the formation of the scales, the boiler plate faces higher temperature outside and lesser temperature inside due to uneven expansion. The water comes suddenly contact with overheated portion and larger amount steam is formed immediately, this results in development of sudden high pressure which may cause explosion of the boiler.

Prevention of scales:

a. If the scale formation is soft it can be removed by a scrapper, wire brush.
b. By giving thermal shocks, by sudden heating and sudden cooling which makes scale brittle and removed by scrubbing with wire brush.
c. If scale is very hard that is formed by $CaCO_3$ can be removed by washing with 5-10% HCl and CaSO4 can be removed with EDTA solution.

Distinguish between sludge and scale.

Sludge	Scale
They are soft, loose and slimy precipitate.	They form hard deposits.
They form non-adherent deposits and can be easily removed.	They stick firmly to the inner surface of the boiler and are very difficult to remove.
They are formed by substances like $CaCl_2$, $MgCl_2$, $MgSO_4$, $MgCO_3$, etc.	They are formed by substances like CaSO4, $Mg(OH)_2$, etc.
They are formed at comparatively colder portions of the boiler.	They are formed at heated portions of the boiler.
They decrease efficiency of boiler but are less dangerous.	They decrease efficiency of boiler and chances of explosion are also there.
They can be removed by blow down operation.	They can't be removed by blow down operation.

CAUSTIC EMBRITTLEMENT

The formation of brittle and in crystalline cracks in the boiler shell is called caustic embrittlement. The main reason for this is the presence of alkali-metal carbonates and bicarbonates in feed water. In lime-soda process, it is likely that, some residual Na_2CO_3 is still present in the softened water. This Na_2CO_3 decomposes to give NaOH and CO2, due to which the boiler water becomes "Caustic Soda".

Na_2CO_3+ H2O → 2NaOH + CO2

The H2O evaporates, the concentration of NaOH increase progressively creating a concentration cell as given below thus dissolving the iron of the boiler as sodium ferrate (Na2FeO2).

(-)Anode: 'Fe' at bents	Conc.NaOH	Dil.NaOH	'Fe' at plane Surface: Cathode (+)

This causes embrittlement of boiler parts such as bends, joints, reverts etc, due to which the boiler gets fail. The iron at plane surfaces surrounded by dilute NaOH becomes cathodic while the iron at bends and joints surrounded by highly concentrated NaOH becomes anodic which consequently decayed or corroded.

Caustic embrittlement can be prevented:

a. By maintaining the pH value of water and neutralization of alkali.
b. By using Sodium Phosphate as softening reagents, in the external treatment of boilers.
c. Caustic embrittlement can also be prevented by adding Tannin or Lignin or Sodium sulphate which prevents the infiltration of caustic-soda solution blocking the hair-cracks.
d. By adding Na_2SO_4 to boiler water. Na_2SO_4 is added to boiler water so that the ratio **Na2SO4Concentration and NaOH concentration is kept as 1:1:,2:1,3:1 in boilers working**

respectively at pressure up to 10,20 and 30 atmosphere.

NaOH Concentration is kept as 1:1:2:1 and 3:1 in boilers working respectively at pressures up to 10, 20 and 30 atmospheres.

Disadvantages of caustic embrittlement:

The cracking or weakening of the boiler metal causes failure of the boiler.

INTERNAL TREATMENT OF WATER

Suitable chemicals are added to the boiler water either to precipitate or to convert the scale into compounds is called internal treatment of the boiler feed water. Internal treatment can be done following types.

1. Calgon conditioning: Involves in adding calgon to boiler water. It prevents the scale and sludge formation by forming soluble complex compound with CaSO4.

Calgon = Sodium hexa meta phosphate = Na2 [Na_4 (PO3)6]

$$Na2\ [Na_4\ (PO_3)_6] \rightarrow 2Na^+ + [Na_4P_6O_{18}]^{-2}\ 2CaSO_4 + [Na_4P_6O_{18}]^{-2} \rightarrow [Ca2\ P6O18]^{-2} + 2Na_2SO_4$$

2. Phosphate conditioning: The addition of sodium phosphate in hard water reacts with the hardness causing agents and gives calcium and magnesium phosphates which are soft and non-adhere and can be removed easily by blow-down operation.

$$3CaCl_2 + 2\ Na_3PO_4 \rightarrow Ca_3(PO_4)_2 + 6NaCl$$
$$3MgSO_4 + 2\ Na_3PO_4 \rightarrow Mg_3(PO4)_2 + 3Na_2SO_4$$

Generally three types of Phosphates are employed.

i. Tri sodium Phosphate (Na3PO4): is too alkaline used for treat to too acidic water.
ii. Di sodium Phosphate (Na2 HPO4): is weakly alkaline used for treat to weakly acidic water.
iii. Sodium dihydrogen Phosphate (Na H2PO4): is too acidic used for treat to too alkaline water.

3. Colloidal conditioning: The addition of organic substances such as Kerosene, tannin, Gel. These substances gets coated over the scale forming precipitates and gives a loose and non-sticky precipitates which can be removed by using blow-down operation.

Priming:

When a boiler is steaming (i.e., producing steam) rapidly, some particles of the liquid water are carried along with the steam. This process of *'wet steam'* formation is known as **'Priming'**.

Causes:

Priming is caused by:

1. The presence of large amount of dissolved salts
2. High steam velocities
3. Sudden boiling
4. Improper boiler design
5. Sudden increase in steam production rate.

Prevention:

1. By fitting mechanical purifiers
2. Avoiding rapid change in steam rate
3. Maintaining low water levels in boilers
4. Effective softening and filtration of boiler feed water
5. Blow down of the boiler.

Foaming:

The phenomenon of formation of *persistent foam or bubbles* on the surface of water inside the boiler, which do not break easily.

Causes:

1. Pure water has very little foaming whereas the water containing dissolved impurities and suspended matter has a greater tendency to produce foam.

2. The presence of large quantity of suspended impurities and oils lowers the surface tension producing foam.

Prevention:

1. Adding antifoaming chemicals like castor oil.

2. Besides castor oil, Gallic acid and tannic acids, corn oil, cotton seed oil, sperm oil, bees wax, etc., are also used as antifoaming agents.

3. Blow down operation of the boiler can prevent foaming.

4. Removing oil form boiler water by adding compounds like sodium aluminate.

Disadvantages of Priming and Foaming:

Priming and Foaming usually occur together. They are objectionable because: -

1. The dissolved salts in boiler water are carried by the wet steam to super heater and turbine blades where they get deposited and reduce the efficiency.

2. The dissolved salts enter the parts of other machinery where steam is used thereby decreasing the life of machinery.

3. The actual height of the water column cannot be judged thereby making the maintenance

of boiler pressure becomes difficult.

Boiler corrosion:

- The decay of boiler material by chemical or electrochemical attack by its environment is known as "**Boiler corrosion**".
- The main reasons for boiler corrosion are:-

Dissolved Oxygen: water contains usually about 8ml of dissolved oxygen per liter at room temperature. Disadvantages of dissolved oxygen:

In presence of prevailing high temperature, the dissolved Oxygen attacks boiler material as follows:

$$2Fe + 2H_2O + O_2 \rightarrow 2Fe(OH)_2$$

$$4Fe(OH)_2 + O_2 \rightarrow 2[Fe_2O_3.2H_2O] \text{ (rust)}$$

Removal of dissolved Oxygen:

1) By adding calculated quantity of Sodium sulphite (Na_2SO_3) or hydrazine(N_2H_4) or sodium sulphide (Na_2S).

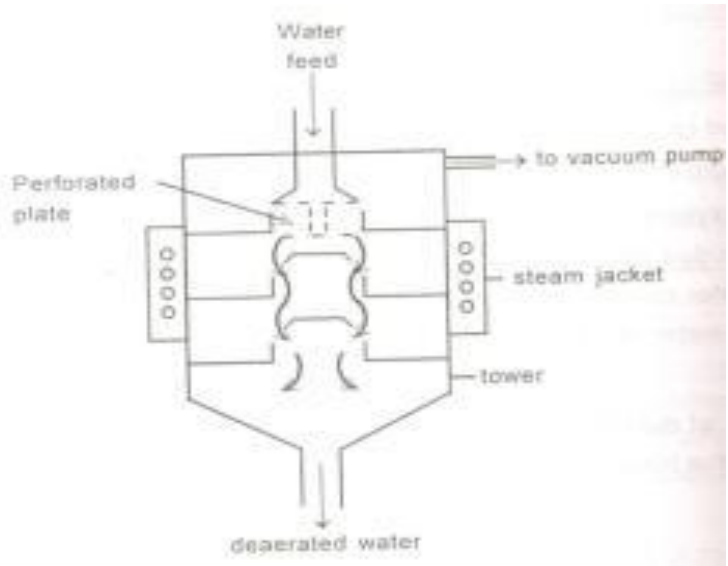
$$2Na_2SO_3 + O_2 \text{ ----------} \blacktriangleright 2\,Na_2SO_4$$

$$N_2H_4 + O2 \quad \text{----------} \quad N_2 + 2H_2O$$

$$Na_2S + 2O2 \quad \text{----------} \blacktriangleright Na_2SO_4$$

Hydrazine is an ideal internal treatment chemical for the removal of dissolved oxygen. Azamina 8001-RD (a polyvalent organic compound) has been employed for degassing water in minimum time.

2) By Mechanical de-aeration i.e., spraying water through a perforated plate fitted in degasification tower, heated from sides and connected to a vacuum pump as shown in the figure.



High temperature, low pressure and large exposed surface (provided by perforated plates) reduce the dissolved oxygen in water.

3) Dissolved Carbon dioxide: has a slow corrosive effect on the materials of boiler plate.

Under the high temperature and pressure the bicarbonates decompose to produce CO2.

$Mg(HCO_3)_2$-------► $MgCO_3 + H_2O + CO_2$

Disadvantages of CO_2:

It has slow corrosive effect on boiler plates by producing the carbonic acid.

$CO_2 + H_2O$---------- ► H_2CO_3

Removal of CO_2:

By adding calculated amount of ammonia.

$2NH_4OH + CO_2 \rightarrow (NH_4)_2CO_3 + H_2O$ By mechanical deaeration process along with oxygen.

Acids from dissolved salts: water containing dissolved Mg salts liberate acids on hydrolysis.

Ex: $MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 + 2HCl$

Disadvantages of acids:

The acids react with the iron of boiler plate in a chain reaction producing HCl again and again.

$Fe+2HCl \rightarrow FeCl_2 + H_2$

$FeCl_2+2H_2O \rightarrow Fe(OH)_2 +2HCl$

Presence of even a small amount of $MgCl_2$ will have corrosion of iron to a large extent.

Prevention of acid corrosion:

1) By softening of boiler water to remove $MgCl_2$ from the water.
2) By frequent blow down operation.
3) Addition of inhibitors to form a thin film on the surface of the boiler protecting the metal from attack. Sodium silicates, sodium phosphates and sodium chromate acts as good corrosion inhibitors.

<u>External Removal Process</u>

1) Ion exchange process is also known as demineralization process. Ion- Exchange resins are insoluble. Cross linked long chain organic polymers with a micro porous structure, and the "functional Groups" attached to the chains are responsible for the ion-exchanging properties. Resins with acidic functional group are capable of exchanging H+ ions with other cations. Resins with basic functional groups are capable of exchanging OH^- ions with other anions.

Resins are classified as:

i. Cation Exchange Resins ii. Anion Exchange Resins.

i. Cation Exchange Resins: Cation exchange resins are styrene divinyl benzene co-polymers, which on sulphonation (or) carboxylation, which contains -COOH, -SO3H functional groups which responsible for exchanging their hydrogen ions with cations in water.

$2RH + Ca(HCO)2 \rightarrow R2Ca + H2CO3$, $2RH + Mg(HCO)2 \rightarrow R2Mg + H2CO3$, $2RH + CaCl2 \rightarrow R2Ca + 2HCl$

ii. Anion Exchange Resins: Anion exchange resins are Phenol formaldehyde (or) amine formaldehyde copolymers, which contains amino or basic functional groups which responsible for exchanging their OH^- ions with anions in water.

$ROH + HCl \rightarrow RCl + H2O$

$2ROH + H2SO4 \rightarrow R2SO4 + 2H2O$

$ROH + H2CO3 \rightarrow RHCO3 + H2O$(ROH = anion exchange resin)

In ion-exchange process, hard water is allowed to pass through cation exchange resins, which remove Ca^{+2} and Mg^{+2} ions and exchange equivalent amount of H^+ ions. Anions exchange resins remove bicarbonates, chlorides and sulphates from water exchange equivalent amount of OH^- ions.

Thus by passing hard water through cation hardness is observed by the following reactions. H^+ and OH^-ions, thus released in water from respective cation and anion exchange columns, get combined to produce water molecules.

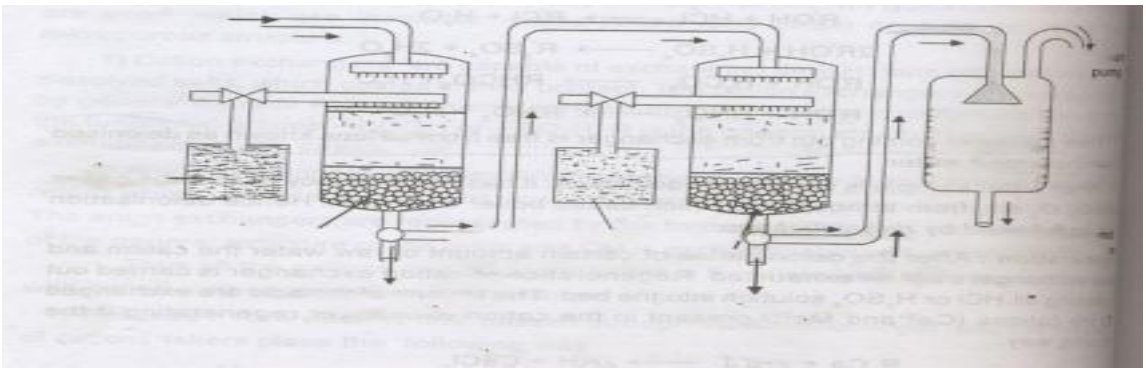
$H+ + OH^- \rightarrow H2O$

The water coming out from the exchanger is ion free from anions and cations. Thus water of zero hardness is obtained.

Regeneration: When cation exchanger losses capacity of producing H+ ions and exchanger losses capacity of producing OH^- ions, they are said to be exhausted. The exhausted cation exchanger is regenerated by passing it through dilute sulphuric acid.

$R2Ca + 2HCl \rightarrow 2RH + CaCl_2$, $R2Mg + 2H2SO4 \rightarrow 2RH + MgSO_4$

The exhausted anion exchanger is regenerated by passing a dilute solution of NaOH.



$RCl + NaOH \rightarrow ROH + NaCl$

$RHCO3 + NaOH \rightarrow ROH + NaHCO_3$

Merits of Ion-exchange process:

- The process can be used to soften highly acidic or alkaline water.
- It produces water of very low hardness (2ppm)
- So it is very good for treating water for use in high-pressure boilers.

Demerits of Ion-exchange process:

- The equipment is costly and more expensive chemicals are needed.
  - If water contains turbidity, the output of the process is reduced. The turbidity must be below 10ppm; else it has to be removed by coagulation and filtration.

2) Lime-Soda process:

In this process, *lime* [$(Ca(OH)_2)$] and *soda* [Na_2CO_3] are the reagents used to precipitate the dissolved
salts of Ca^{+2} and Mg^{+2} as $CaCO_3$ and $Mg(OH)_2$ which are later on filtered off.

Lime reacts with temporary hardness causing salts, Mg permanent hardness, CO_2, acids, bicarbonates and alums.

Lime cannot remove permanent hardness which should be removed by soda.

Precautions to be followed during process:

Only calculated amounts of lime and soda must be added. Excess of lime-soda causes boiler troubles like corrosion and caustic embrittlement.

The chemical reactions of lime and soda are slow and the precipitates $CaCO_3$ and $Mg(OH)_2$ are very fine.

Proper time must be given for the reactions of softening process to complete, because all the reactions of the process are slow.

Cold lime soda process:

Calculated amounts of lime and soda are mixed with water at room temperature. Cold lime soda process provides water containing residual hardness of 50-60 ppm.
The precipitates formed are finely divided which do not settle down and can"t be filtered easily.
So, it is essential to add small amounts of coagulants (like alum, aluminium sulphate, sodium aluminate) which hydrolyse to flocculent, gelatinous precipitate of $Al(OH)_3$ and entraps the precipitate.

Use of sodium aluminate as **Coagulant** also helps in the removal of silica as well as oil present

in the water.

$NaAlO_2 + 2H_2O \rightarrow NaOH + Al(OH)_3$

$Al_2(SO_4)_3 + 3Ca(HCO_3)_2 \rightarrow 2Al(OH)_3 \downarrow + 3CaSO_4 + 6CO_2$

Method:

Raw water and calculated quantities of chemicals (lime+soda+coagulant) are fed from the top into inner vertical circular chamber fitted with a vertical rotating shaft carrying a number of paddles.

As the raw water and chemicals flow down, there is vigorous stirring and continuous mixing whereby softening of water takes place.

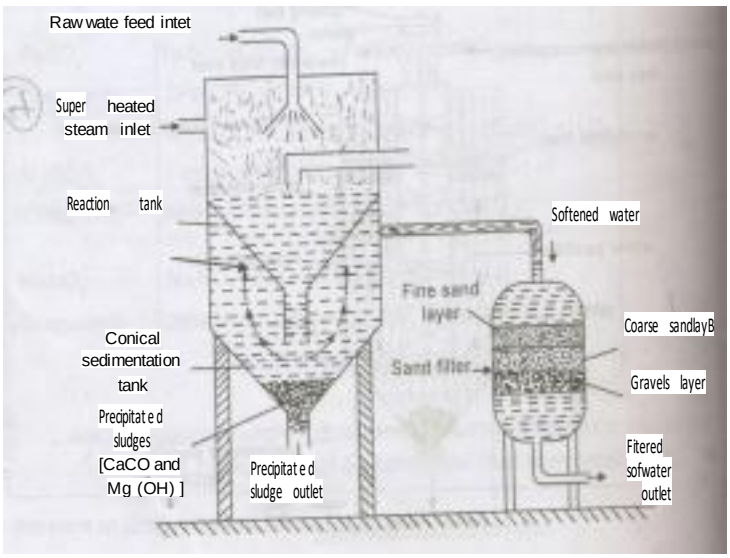
The heavy sludge settles down in the outer chamber by the time softened water reaches up.

The softened water then passes through a filtering media (made of wood fibres) to ensure complete removal of sludge.

Filtered soft water finally flows out continuously through the outlet at the top and sludge settling at the bottom of the outer chamber is drawn off occasionally.

Hot Soda Lime Process:

It involves treating water with softening chemicals at a temperature of 80^0-150^0 C. Since hot process is operated at a temperature close to the boiling point of the solution, so



a) The reaction proceeds faster,

b) Softening capacity of hot process is increased,

c) The precipitate and sludge formed settle down rapidly and hence no coagulants are needed,

d) Much of the dissolved gases (CO2) are driven out,

e) Viscosity of softened water is lower so filtration of water becomes much

Hot lime soda process produces water of residual hardness of 15-30ppm.

Process:

It contains essentially three parts

1) A reaction tank in which water, chemical and steam are thoroughly mixed.
2) A conical sedimentation vessel in which sludge settle down.
A sand filter which ensures complete removal of sludge from the softened water.

Advantages of Lime-Soda process:

1) It is very economical.
2) The process increases the pH value of treated water thereby corrosion of distribution pipes is reduced.
3) Besides removal of hardness causing salts, minerals, iron and manganese present in water is removed.
4) Due to alkaline nature, amount of pathogenic bacteria in water is considerably reduced.

Disadvantages Of Lime-Soda Process:

1) Skilled supervision is required for efficient and economical softening.
2) Disposal of large amounts of sludge poses a problem.
3) This can remove hardness only up to 15ppm, which is not good for boilers.

Differences between cold and hot lime soda process: -

Sl.No.	Cold Lime Soda Process	Hot Lime Soda Process
1.	It is done at room temperature.	It is done at elevated temperatures (84-100 ^{0}C).
2.	It is slow process.	It is rapid process.
3.	Use of coagulants is must.	Coagulants are not needed.
4.	Filtration is not easy.	Filtration is easy as viscosity of water becomes low at elevated temperatures.
5.	Softened water has residual hardness around 60 ppm.	Softened water has residual hardness around 15-30 ppm.
6.	Dissolved gases are not removed.	Dissolved gases such as CO_2 are removed to some extent.
7.	It has low softening capacity.	It has high softening capacity.

Points to be followed while solving problems in lime soda problems:

1) To convert the hardness of the hardness causing salts in terms of $CaCO_3$ equivalents.

2) If $CaCO_3$ is given as hardness causing salt, it should also be considered as temporary hardness expressed in terms of $CaCO_3$.

3) $MgCO_3$, $Mg(HCO_3)_2$, should be converted to $CaCO_3$ equivalents.

4) Some salts like NaCl, KCl, Silica, K_2SO_4, Fe_2O_3 neither cause hardness nor react with lime and soda. Hence they must not be considered for calculation of lime and soda required.

5) The hardness of $Al_2(SO_4)_3$ must be multiplied by 3 for both lime and soda, HCl hardness must be multiplied by 0.5, $Mg(HCO_3)_2$ hardness must be multiplied by 2, $NaHCO_3$, $KHCO_3$, $NaAlO_2$ hardness
multiplied by 0.5 and should be subtracted for calculation of amount of lime and soda required respectively.

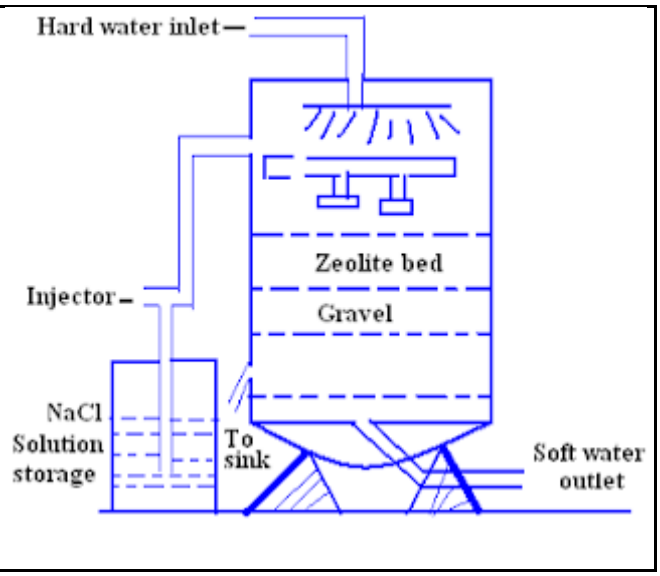
3) <u>Zeolite Process</u>

Zeolite are hydrated sodium alumino siticates $Na_2O.Al_2O_3\ xSiO_2.yH_2O$. (x=2-10 and y=2-6) (inorganic salts). They work as water softners by replacing the calcium and magnesium ions in water with the sodium ions in zeolite. Zeolite Natural zeolites Synthetic zeolites 1. Natural zeolites Non-porous, amoamorphous and durable e.g. $Na_2O.\ Al_2O_3.3SiO_2.2H_2O$ (Natrolite) $CaO.Al_2O_3.4SiO_2.4H_2O$ (Laumonite) 2. Synthetic zeolites Gel like structures, generally porous Na_2CO_3, Al_2O_3 and SiO_2 heated together $Na_2O.Al_2O_3\ xSiO_2.yH_2O$. (x=2-10 and y=2-6)

For softening of water by zeolite process, hard water is percolated at a specified rate through a bed of zeolite. Zeolite holds sodium ion loosely and can be represented as Na_2Z, where Z represents insoluble radical frame work.

When the water passes through the zeolite the hardness causing ions (Ca^{+2}, Mg^{+2} etc.) are retained by the zeolite as CaZ and MgZ respectively, while the outgoing water contains equivalent amount of sodium salts. The block diagram and chemical reactions taking place in zeolite softener are:

$$Ca(HCO_3)_2 + Na_2Z \rightarrow CaZ + 2NaHCO_3$$
$$Mg(HCO_3)_2 + Na_2Z \rightarrow MgZ + 2NaHCO_3$$
$$CaSO_4 + Na_2Z \rightarrow CaZ + Na_2SO_4$$
$$CaCl_2 + Na_2Z \rightarrow CaZ + 2NaCl$$
$$MgSO_4 + Na_2Z \rightarrow MgZ + Na_2SO_4$$
$$MgCl_2 + Na_2Z \rightarrow MgZ + 2NaCl$$



<u>Regeneration of Zeolite Bed</u>

After some time, when the zeolite is completely changed into calcium and magnesium zeolites, then it gets exhausted (saturated with Ca+2Ca+2 and Mg+2Mg+2 ions) and it ceases

to soften water. It can be regenerated and reused by treating it with a 10% brine (sodium chloride) solution. Regeneration reactions are

$CaZ+2NaCl \rightarrow Na2Z+CaCl$ $MgZ+2NaCl \rightarrow Na2Z+MgCl2$

Advantages:

1. Almost complete removal of hardness (10ppm) 2. It is compact 3. Requires only less time for softening 4. No sludge formation since no precipitate is formed 5. Can work under pressure also

Disadvantages

1. More sodium salt concentration in softened water. 2. Turbidity containing water cannot be used 3. Process exchange only Ca^{2+} and Mg^{2+} ions but cannot exchange HCO_3^- and CO_3^{2-} ions. So cannot be used in boilers 4. If Fe^{2+} and Mn^{2+} are present in large quantities. They form respective zeolites so zeolites cannot be regerated 5. Water consisting of high alkalinity or acidity cannot be used because zeolite is decomposed.

Questions

1. How temporary hardness differs from permanent hardness?
2. Define hardness and estimate hardness of water by using EDTA solution?
3. Explain Caustic embrittlement, Scale & Sludge.
4. Explain determination of fluoride ions (F^-) by ion selective electrode method.
5. Explain softening of water using Ion exchange process.
6. What is desalination? Explain reverse osmosis process.
7. Explain Steps involved in treatment of sewage water.
8. What is Potable water and what are its specifications.
9. Explain Steps involved in the treatment of potable water.
10. 0.28 grams of CaCO3 were dissolved in HCl and the solution was made upto one litre with distilled water. 100 ml of the above solution required 28 ml of EDTA solution on titration. 100 ml of hard water sample consumed 33 ml of same EDTA solution EBT indicator. 100 ml of this water after boiling cooling and filtering required 10 ml of EDTA solution in titration. Calculate the permanent and temporary hardness of water sample in ppm.

11. Explain the following?
    a. Calgon conditioning and Phosphate conditioning of boiler feed water.
    b. What is defluoridation and explain Nalgonda technique.

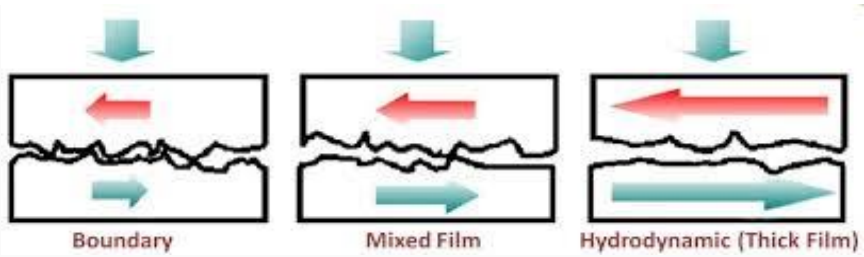
12. What is Hardness and calculate a sample of water is found to contains following analytical data in milligrams per litre Mg(HCO3)2 = 14.6, MgCh = 9.5, MgSO4 = 6.0 and Ca(HCO3)2 = 16.2. Calculate temporary and permanent hardness of water in parts per million, Degree Clarke's and Degree French.

Module – 2: LUBRICATION AND LUBRICANTS

A substance which is capable of reducing friction between two surfaces which are sliding over each other is called Lubricant. Due to Lubricant the co-efficient of friction between two rubbing surfaces is reduced. The loss of energy due to friction is considerably reduced in this manner. Petroleum is the main source of Lubricants, but synthetic Lubricants have also been prepared for specific purpose. The lubricant acts in a number of manners. It acts as a coolant by removing the heat of friction generated as a result of rubbing of surfaces. In lubrication it does not allow a direct contact between the rubbing surfaces. A thin film of lubricant lies between these rubbing surfaces under proper condition of lubrication. In internal combustion engines, it acts as a seal by sealing the piston & cylinder wall at compression rings. Hence there is no leakage of gases at high pressure in the combustion chamber. Lubrication also avoids power loss in the IC engine.

The main features of lubrication are:

1. It reduces wear and tear and surface deformation, by avoiding direct contact between the rubbing surfaces.
2. It reduces the loss of energy in the form of heat by acting as a coolant.
3. It reduces the efficiency of the machine by reducing the waste of energy.
4. It reduces expansion of metal by local frictional heat.
5. It minimizes the liberation of frictional heat and hence avoids seizure of moving surfaces.
6. It prevents unsmooth motion relative motion of the moving or sliding parts.
7. It reduces the maintenances as well as running cost of the machine to a large extent.
8. As seen above it also acts as a seal in IC engines.



CLASSIFICATION OF LUBRICANTS

Lubricants can be broadly classified, on the basis of their physical state, as follows: (1) Liquid lubricants or lubricating oils; (2) Semi-solid lubricants or greases, and (3) Solid lubricants.

1. Lubricating oils

Lubricating oils reduce friction and wear between two moving/sliding metallic surfaces by

providing a continuous fluid film in-between them. They also act as: (a) cooling medium; (b) sealing agent, and (c) corrosion preventer. Good lubricating oil must possess: (a) low pressure (or high boiling point), (b) adequate viscosity for particular service conditions, (c) low freezing point, (d) high oxidation resistance. (e) Heat stability, (f) non-corrosive properties, (g) stability to decomposition at the operating temperatures. Lubricating oils are further classified as:

(1) Animal and vegetable oils: Before the advent of the petroleum industry, oils of the vegetable and animal origins were the most commonly used lubricants. They possess good oiliness (a property by virtue of which the oil sticks to the surface of machine parts, even under high temperatures and heavy loads). However, they: (i) are costly, (ii) undergo oxidation easily forming gummy and acidic products and get thickened on coming in contact with air, (iii) have some tendency to hydrolyze, when allowed to remain in contact with moist-air or aqueous medium. So at present, they are rarely used as such. Actually, they are used as "blending agent" with other ' lubricating oils (like mineral oils) to produce desired effects in the latter.

(2) Mineral or petroleum oils are obtained by distillation of petroleum. The length of the hydrocarbon chain in petroleum oils varies between about 12to 50 carbon atoms. The shorter-chain oils have lower viscosity than the longer-chain hydrocarbons. These are the most widely used lubricants, because they are; (i) cheap, (ii) available in abundance, and (iii) quite stable under service conditions. However, they possess poor oiliness as compared to that of animal and vegetable oils. Tile oiliness of petroleum oils can be increased by the addition of high molecular weight compounds like oleic acid, stearic acid, etc.

Purification: Crude liquid petroleum oils contain lot of impurities (like wax, asphalt,etc.) and consequently, they have to be thoroughly purified before being put to use.(i) The wax, if not removed, raises the pour-point and renders the lubricating oil unfit for use at low temperatures. (ii) Certain constituents get easily oxidized under working conditions and cause sludge formation. (iii) Some constituents mainly asphalt, undergo decomposition at higher temperatures, causing carbon deposition and sludge formation. A number of processes are used for removing these unwanted impurities by using Dewaxing or acid refining or by solvent refining.

(3) Blended oils: No single oil saves as the most satisfactory lubricant for many of the modern machineries. Typical properties of petroleum oils are improved by incorporating specific additives. These so-called 'blended oils' give desired lubricating properties, required for particular machinery. The following additives are employed

(i) Oiliness-carriers: Oiliness of a lubricant can be increased by addition of an oiliness-carrier like vegetable oils (e.g., coconut oil, castor oil) and fatty acids (like palmitic acid, stearic acid, oleic acid. etc.).

(ii) Extreme-pressure additives: Under extreme-pressure, a thick film of oil is difficult to maintain, and the oil need to have a high oiliness. Besides improving oiliness directly, high-pressure additives are used. these additives contain certain materials which are absorbe d on the metal surface or react chemically with metal, producing a surface a layer of low shear-strength on the metal surface, thereby preventing the tearing up of the metal. Another property of high-pressure additives is that they react, at high temperature on metal surfaces, forming surface alloys so as to prevent the welding together of the rubbing parts under severe operating conditions.

The main substances added for high-pressure lubrication are :(a) fatty ester, acids, etc., which form oxide film with the metal surface; (i) organic materials, which contain Sulphur; (c) organic chlorine compounds; (d) organic phosphorus compounds. High-pressure lubricants also contain some lead in order to produce thin film of lead sulphide and other lead compounds on the surfaces of machines like gear teeth.

(iii) Pour-point depressing additives used are phenol and certain condensation products of chlorinated wax with naphthalene. These prevent the separation of wax from the oil.

(iv) Viscosity-index improvers are certain high molecular weight compounds like hexanol.

(v) Thickeners such as polystyrene are materials usually of molecular weight between 300 and 3,000. They are added in order to give the lubricating oil a higher viscosity.

(vi) Antioxidants or inhibitors, when added to oil, retard oxidation of oil by getting them-selves preferentially oxidized. They are particularly added in lubricants used in internal combustion engines, turbines, etc., where oxidation of oil is a serious problem. The antioxidants are aromatic, phenolic or amino compounds.

(vii) Corrosion preventers are organic compounds of phosphorus or antimony. They protect the metal from corrosion by preventing contact between the metal surfaces and the corrosive substances.

(viii) Abrasion inhibitors like tricresyl phosphate.

(ix) Antifoaming agents (like glycols and glycerol) help in decreasing foam formation.

(x) Emulsifiers such as sodium salts of sulphonic acid.

(xi) Deposit inhibitors are detergents such as the salts of phenol and carboxylic acids . Deposits are formed in internal combustion engine, due to imperfect combustion. Such additive disperses and cleans the deposits.

2. GREASES OR SEMI-SOLID LUBRICANTS

Lubricating grease is a semi - solid, consisting of a soap dispersed throughout liquid lubricating oil. The liquid lubricant may be petroleum oil or even synthetic oil and it may contain any of the additives for specific requirements. Greases are prepared by saponification of fat (such as tallow or fatty acid) with alkali (like lime, caustic soda, etc.), followed by adding hot lubricating oil while under agitation. The total amount of mineral oil added determines the consistency of the finished grease. The structure of lubricating greases is that of a gel. Soaps are gelling agents, which give an interconnected structure (held together by intermolecular forces) containing the added oil. At high temperatures, the soap dissolves in the oil, whereupon the interconnected structures cease to exist and the grease liquefies. Consistency of greases may vary from a heavy viscous liquid to the of a stiff solid mass. To improve the heat-resistance of grease, inorganic solid thickening agents (like finely divided clay, bentonite, colloidal silica, carbon black, etc.) are added.

Greases have higher shear or frictional resistance than oils and, therefore, can support much heavier loads at lower speeds. They also do not require as much attention unlike the lubricating liquids. But greases have a tendency to separate into oils and soaps. Grease are used : (i) in situations where oil cannot remain in place, due to high load, low speed, intermittent operation, sudden jerks, etc. e.g. rail axle boxes, (ii) in bearing and gears that work at high temperatures ; (iii) in situations where bearing needs to be sealed against entry of dust, dirt, grit or moisture, because greases are less liable to contamination by these ; (iv) in situations where dripping or spurting of oil is undesirable, because unlike oils, greases if used do not splash or drip over articles being prepared by the machine. For example, in machines preparing paper, textiles, edible articles, etc.

The main function of soap is thickening agent so that grease sticks firmly to the metal surfaces. However, the nature of the soap decides: (a) the temperature up to which the grease can be used; (b) its consistency; (c) Its water and oxidation resistance. So, greases are classified after the soap used in their manufacture. Important greases are: (i) Calcium-based greases or cup-greases are emulsions of petroleum oils with calcium soaps. They are, generally, prepared by adding requisite amount of calcium hydroxide to hot oil (like tallow) while under agitation. These greases are the cheapest and most commonly used. They are insoluble in water, so water resistant. However, they are satisfactory for use at low temperatures, because above 80°C, oil and soap begins to separate out.

(ii) Soda-base greases are petroleum oils, thickened by mixing sodium soaps. They are not water resistant, because the sodium soap content is soluble in water. However, they can be used up to 175°C. They are suitable for use in ball bearings, where the lubricant gets heated due to friction.

(iii) Lithium-based greases are petroleum oils, thickened by mixing lithium soaps. They are water-resistant and suitable for use at low temperatures [up to 15°C] only.

(iv) Axle greases are very cheap resin greases, prepared by adding lime (or any heavy metal hydroxide) to resin and fatty oils. The mixture is thoroughly mixed and allowed to stand, when grease floats as stiff mass. Filters (like talc and mica) are also added to them. They are water-resistant and suitable for less delicate equipments working under high loads and at low speeds. Besides the above, there are greases prepared by dispersing solids (like graphite, soapstone) in mineral oil. These are mostly used in rail axle boxes, machine bearings, tractors rollers, wires ropes etc.

3. SOLID LUBRICANTS

Solid lubricants are used where: (i) operating conditions are such that a lubricating film cannot be secured by use of lubricating oils or greases; (ii) contamination (by the entry of dust or grit particles) of lubricating oil or grease is unacceptable, (iii) the operating temperatures or load is too high even for a semi-solid lubricant to remain in position; and (iv) combustible lubricants must be avoided.

The two most usual solid lubricants employed are graphite and molybdenum disulphide. Graphite consists of a multitude of flat plates, one atom thick, which are held together by only weak bonds, so that the force to shear the crystals parallel to the layers is low. Consequently, the parallel layers slide over one another easily. Usually, some organic substances are mixed solid lubricants so that they may stick firmly to the metal surface.

On the other hand, molybdenum disulphide has a sandwich like structure in which a layer of a Mo atoms lies between two layers of S atoms. Poor interlaminar attraction is responsible for low shear strength in a direction parallel to the layers. Solids lubricants are used either in the dry powder or mixed with water or oil. The solids fill up the low spots in the surfaces of moving parts and form solid films, which have low frictional resistance. The usual coefficient of friction between solid lubricants is between 0.005 and 0.01.

(a) Graphite is the most widely used of all solid lubricants. It is very soapy to touch, non-inflammable and not oxidized in air below 375°C. In the absence of air, it can be used upto very much higher temperatures. Graphite is used either in powdered form or as suspension. Suspension of graphite in oil or water is brought about with the help of an emulsifying agent like tannin. When graphite is dispersed in oil, it is called 'oildag' and when it is dispersed in water; it is called 'aquadag'. Oildag is found particularly useful in internal combustion engines, because it forms a film between the piston rings and the cylinder and gives a tight-fit contact, thereby increasing compression. On the other hand, oildag is useful where a lubricant free from oil is needed. e.g., foodstuffs industry. Graphite is also mixed with greases to form graphite-greases, which are used at still higher temperatures.

Uses: As lubricant in air-compressors, lathes, general machine-shop works, foodstuffs industry, railway track-joints, open gears, chains, cast iron bearings, internal combustion engine, etc. (b) Molybdenum disulphide possesses very low coefficient of friction and is stable in air up to 400°C. Its fine powder may be sprinkled on surfaces sliding at high velocities, when it fills low spots in metal surfaces, forming its film. It is also used along with solvents and in greases. Besides the more important graphite and molybdenum disulphide, other substances like soapstone, talc, mica, etc., are also used as solid lubricants.

4. SYNTHETIC LUBRICANTS

Modern synthetic lubricants possess, in general, the following distinguishing characteristics: (1) non-inflammable, (ii) high flash points, (iii) high thermal stability at high operating temperatures, (iv) high viscosity-index, (v) chemical stability, etc. Important synthetic lubricants are given below.

(1) Polymerized hydrocarbons like polyethylene, polypropylene, polybutylene in the molecular weight range of 500 to 50,000 are residue-free, light in color, free from non-hydrocarbon impurities, chemically non-reactive and high temperature lubricants.

(2) Polyglycols and related compounds like polyethylene glycol, polypropylene glycol, polyglycidyl ethers, and higher polyalkylene oxides can be used as water-soluble as well as water-insoluble lubricants in rubber bearings and joints. Polyglycidyl ethers and higher polyalkylene oxides are water-insoluble, but they can absorb a considerable amount of water.

(3) Organic amines, imines and amides are good synthetic lubricants, since they possess low pour-points and high viscosity-index. They can be used under temperature conditions of -50°C to 250°C

(4) Silicones are very good synthetic lubricants, because are not oxidized below 200°C and possess high viscosity-index.

(5) Fluorocarbons are not decomposed by heat, not easily oxidizable and chemically inert and resistant to chemicals, except molten sodium.

LUBRICATING EMULSIONS

An emulsion is two-phase system, consisting of a fairly coarse dispersion of two immiscible liquids, the one being dispersed as fine droplets in the other. The disperse (or the internal) phase is the liquid that is broken into droplets. The surrounding liquid is known as the continuous or external or dispersing phase. Usually, the size of dispersed droplets varies from 1 to 6 micron. A dispersion system consisting of two immiscible liquids is inherently unstable, and to increase its stability, a third agent, called emulsifier or emulsifying agent, is added. Emulsifiers are compounds exhibiting both polar and non-polar character. The emulsifier molecule contains a hydrophobic-end and a hydrophilic end. Hydrophobic-end of

the molecule is preferably wetted by oil; whereas the hydrophilic-end is wetted by water. Thus, emulsifier molecule is adsorbed at the interface of the two phases (oil and water), resulting in the formation of a protective film around the dispersed droplets. A Sodium Soap molecule illustrates well the functioning of an emulsifier. The sodium soap is the sodium salt of a long chain fatty acids like $C_{15}H_{31}COONa$, possessing a hydrophilic group -COONa and hydrophobic-end, $C_{15}H_{31}$.

Geometric calculations have shown that the maximum amount of dispersed phase in the other liquid can be 74.02% of the total volume. Thus, a water-oil emulsion mixture with less than 26% oil would tend to form an O/W emulsion; whereas a mixture with more than 74% oil would result in a W/O emulsion. Composition between 26 and 74% can result in both types of emulsion.

(a) Oil in-water emulsions are obtained by adding oil containing about 3-20% water-soluble emulsifying agent to a suitable quantity of wafer. The most usual emulsifying agents are sodium soaps and sodium and potassium salts of sulphonic acids. The main use of such an emulsion is as cooling and lubricating liquid for cutting tools. Another use is lubrication for certain rather heavy sliding components such as pistons in marine diesel engines and large internal combustion engines. Such emulsions also give rust protection.

(b) Water-in-oil emulsions are prepared by 'mixing together water containing about 1 to 10 % water- soluble emulsifier (Like alkaline-earth soap, e.g., calcium stearate) with oil. These emulsions possess much higher viscosity than that of the oil from which they are prepared. An emulsion which uses about 40% water by volume is widely used to lubricate compressors and pneumatic tools. They provide cooling effect (due to the evaporation of water), besides lubrication action.

Characteristics:

A good lubricant generally possesses the following characteristics:

- A high boiling point and low freezing point (in order to stay liquid within a wide range of temperature)
- A high viscosity index.
- Thermal stability.
- Hydraulic stability.
- Demulsibility.
- Corrosion prevention.
- A high resistance to oxidation.
- Anti scoring property
- Water antiaffinity property
- Anti foaming property

Anti scoring property: Lubricants should have the property to prevent the scoring failure of tooth surface under high pressure of load.

Water anti affinity property: Moisture tends to condense due to temp. change when the gears are stopped. Lubricants should have the property of isolating moisture and water from it.

Anti foaming property: Lubricants should form no foam under agitation.

More Discussion on property:

VISCOSITY INDEX : Viscosity index is an expression of effect of change of temperature on the viscosity of oils. This change can be evaluated numerically and the result is expressed as VI. It Is the avg. Decrease in viscosity of oil per degree rise in temp between 100Of AND 2100F. Silicones, poly glycol ethers, diesters and trimesters have high VI.

Drop Point: It is that temp at which grease passes from the semi liquid to liquid state. It determines the upper temp limit for the application of grease.

FLASH POINT: Flash point is the temperature at which the oil gives off sufficient vapors which can be ignited. It reflects on the capability of the oil to work at higher temperature without any fire hazard.

Cloud point It is the temperature below which wax in diesel forms a cloudy appearance. In general it is that temp at which the lubricant becomes cloudy or hazy when cooled. And that waxes thickens the oil and clogs fuel filters and injectors in engines.

pour point is the minimum temperature at which a lubricants turns into semi solid and almost losses its flow characteristic. Both indicate suitability of lubricant in cold conditions and must be low. It reflects on the capability of the oil to work at low temperatures.

Iodine Number: It is the no. Of gm. equivalent of I2 to the amount of ICL absorbed by 100 gm of oil. Each oil has specific I N. It determines the extent of contamination of oil/ amount of un saturation in oil. Low IN is good for oil.ICL—Wijs solution. RCH=CHR + ICl →RCHI-CH(Cl)R

IN	Oil Type	Example
>150	drying	linseed, tung oil

100-150 semidrying castor, soyabeen

<100 non-drying olive, cocanut

Aniline point: It is the minimum temp at which oil is miscible with equal amount of aniline. It is a measure of aromatic content of an oil. It is an indication of possible determination of rubber sealing.

Aniline + sample oil → Homogeneous solution → cloudiness

Emulsion Number: It is the property of water to get mixed with oil easily or the reverse. A good lubricating oil should form such an emulsion with water which breaks easily. This is known as demulsification. The time in seconds in which a given volume of oils and water separates out into distinct layers is called demulsification number. A good lubricating oil should have low demulsification number. In cutting oils the higher the emulsification number, better the oil is. Emulsion acts both as lubricant & coolant. It is of the following types.

OIW – (26% O+ 74% W) ---- Sodium& potassium salt of sulphuric acid as dispersed phase and water as medium. Milk, Vanishing cream

WIO--- (74% O+26% W)----- water as dispersed phase & oil as dispersion medium (Butter, cold cream)

Lubrication Mechanism:

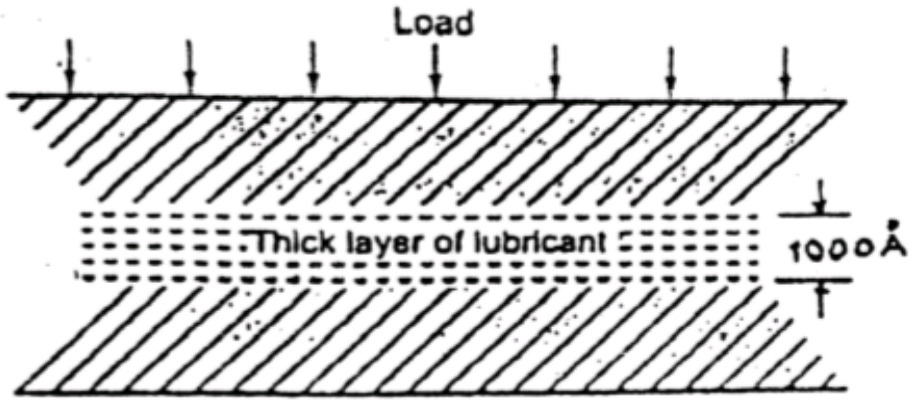
Thick Film Mechanism: In this case the sliding surfaces are separated completely by applying a thin uniform film of liquid lubricants between them.

The thickness of liquid film is at least 1000 Armstrong. The liquid lubricants do not have any chemical affinity to the metal surface and it sticks to it due to viscosity.

The liquid film covers all irregularities in the sliding surface and thus prevents formation of welded junction and hence avoids contact between sliding surface.

The liquid lubricant must have between sliding surface. The liquid lubricant must have sufficient viscosity so as to maintain the fluid film in its place. If viscosity is higher, a large amount of energy is required.

When the load is applied, the pressure developed in the lubricant is sufficient to keep moving surfaces apart and hence it known as hydrodynamic lubrication. It occurs in journal bearing. This type of lubrication is used in watches, clocks, electric motors, turbines etc.



<u>Thin Film/Boundary Lubrication</u> : Boundary lubrication exists when the operating condition are such that it is not possible to establish a full fluid condition, particularly at low relative speeds between the moving or sliding surfaces.

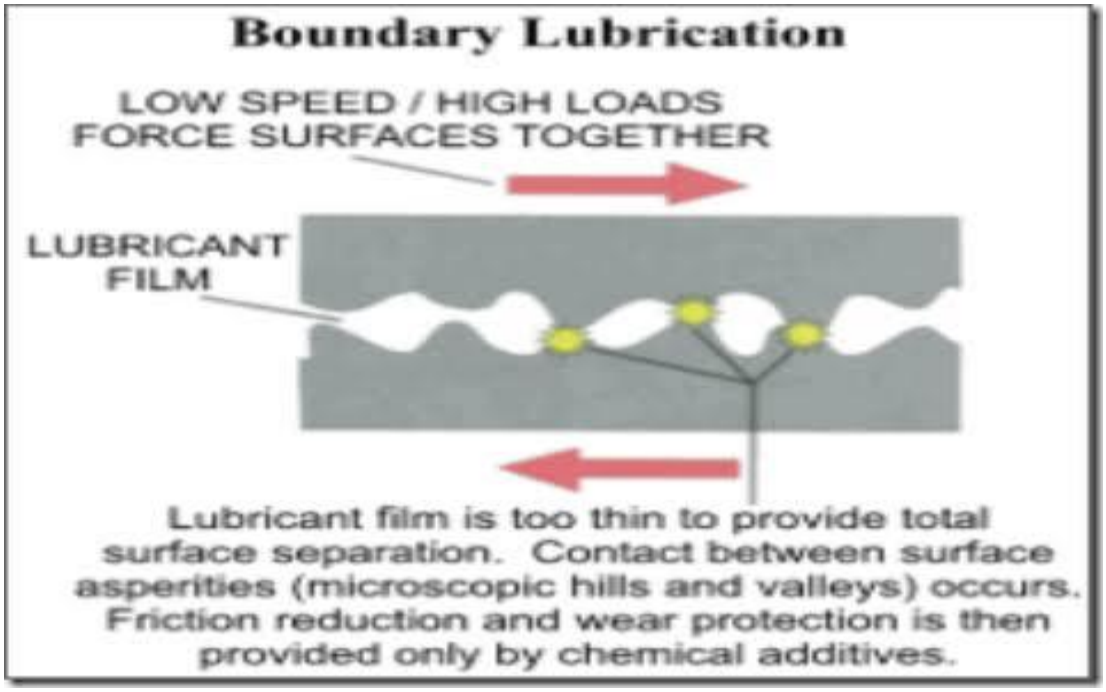
The oil film thickness may be reduced to such a degree that metal to metal contact occurs between the moving surfaces. The oil film thickness is so small that oiliness becomes predominant for boundary lubrication.

Boundary lubrication happens when

- A shaft starts moving from rest.
- The speed is very low.
- The load is very high.
- Viscosity of the lubricant is too low.

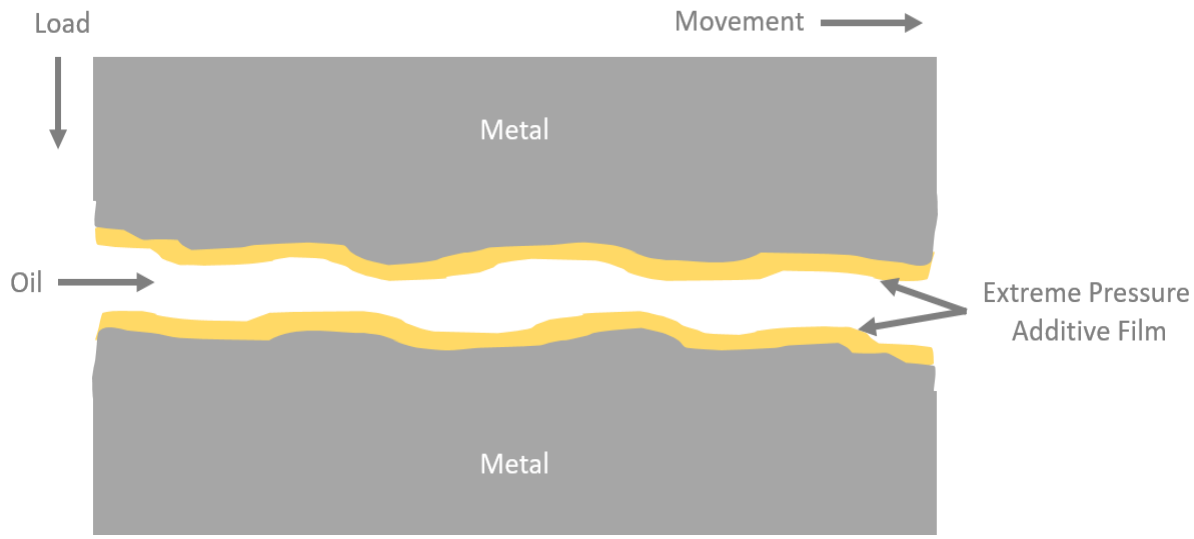
<u>Examples for boundary lubrication:</u>

- Guide and guide shoe in two stroke engine.
- Lubrication of the journal bearing in diesel engines (mainly during starting and stopping of engine).
- Piston rings and when cylinder liner is at TDC and BDC position when the piston direction changes and if the relative speed is very slow.



Extreme pressure Lubrication

When the moving or sliding surfaces are under very high pressure and speed, a high local temperature is attained. Under such condition, liquid lubricant fails to stick to the moving parts and may decompose and even vaporize. To meet this extreme pressure condition, special additives are added to the minerals oils. These are called “extreme pressure lubrication." These additives form on the metal surfaces more durable films capable of withstanding high loads and high temperature. Additives are organic compounds like chlorine (as in chlorinated esters), sulphur (as in sulphurized oils), and phosphorus (as in tricresyl phosphate).



Module – 3: CORROSION

Corrosion is the gradual deterioration of a metal from its surface due to the chemical or electrochemical interaction with a corrosive environment. It is a diffusion- controlled process so it occurs on exposed surfaces.

Types of corrosion: 1) Chemical or Dry

2) Electrochemical or wet

Dry corrosion: By oxidation, By other gases, By liquid metal

A) Oxidation: Due to direct oxidation of oxygen on metals in the absence of moisture.

$2M + nO_2 \rightarrow 2M^{n+} + 2nO^{2-} \rightarrow Mn_2O_3$

At low temp. all alkali and alkaline metals are oxidised easily where as at high temp. except Ag, Au and Pt, all other metals are oxidised.

B) Liquid- Metal corrosion: By passing a liquid metal over a solid metal at high temp. which leads to the weakening of the metal surface either due to its dissolution in liquid metal or penetration of liquid metal in to the solid surface. EX) Coolant (Na metal) leads to corrosion of cadmium in a nuclear reactor.

C) Corrosion by other gases : Gases like chlorine and H_2S show corrosive effect on a metal surface due to their chemical affinity.

$2Ag + Cl_2 \rightarrow 2AgCl$ (AgCl coating is protective)

$Sn + 2Cl_2 \rightarrow SnCl_4$ ($SnCl_4$ is volatile and hence non-protective)

Electrochemical Corrosion: It is mostly through the formation of galvanic cells. It involves mostly separate anode and cathode between which the current flows/ corrosion at anode generates metal ions/ non-metallic ions like OH^- & O^{2-} at cathode./ diffusion of ions through the conducting medium/ a single metal partly behaving as anode and cathode due to differential aeration.

EX) Rusting of iron :

Reactions: Oxidation : $Fe - 2e \rightarrow Fe^{2+}$

Reduction :

a) Oxygen Absorption $4H^+ + O_2 + 4e \rightarrow 2H_2O$ (Acidic medium)

$2H_2O + O_2 + 4e \rightarrow 4OH^-$ (neutral medium)

b) Hydrogen Evolution(Absence of oxygen) :

$$2H^{+} + 2e \rightarrow H_2 \text{ (Acidic medium)}$$

$$2H_2O + 2e \rightarrow H_2 + 2OH^{-} \text{ (neutral or alkaline)}$$

Final Reaction : $Fe^{2+} + 2OH^{-} \rightarrow Fe(OH)2$. This in presence of excess oxygen oxidised to Fe(OH)3.

Which is RUST, $Fe_2O_3.3H_2O$.

Pilling Bedworth Rule:

Ratio of the volume of the oxide film/ coating formed to the volume of the metal consumed.
P. B Ratio = volume of metal oxide/ volume of metal
Also known as specific volume, it carries significance for determining rate of corrosion.
case-1 → volume of oxide > volume of metal, PBR > 1, Protective (Ti, Al, Cr, Cu)
case-2 → volume of oxide < volume of metal, PBR < 1, Non protective (Li, Na, K, Mg)
case-3 → volume of oxide >> volume of metal, PBR > 2, Non protective (Mo, W, Nb)

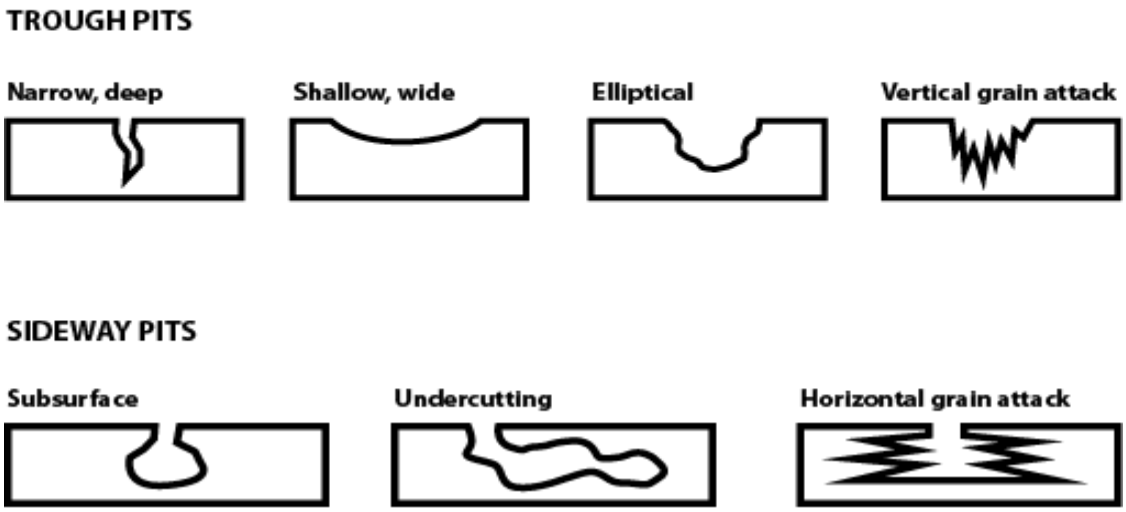
Metal	PBR
Pb	1.4
Ni	1.52
Cr	1.99
Cu	1.68
K	0.45
Li	0.78
Nb	2.61

TYPES OF ELECTROCHEMICAL CORROSION

PITTING CORROSION

Pitting is one of the most destructive types of corrosion, as it can be hard to predict, detect and characterize. Pitting is a localized form of corrosion, in which either a local anodic point, or more commonly a cathodic point, forms a small corrosion cell with the surrounding normal surface. Once a pit has initiated, it grows into a “hole” or “cavity” that takes on one of a variety of different shapes. Pits typically penetrate from the surface downward in a vertical direction. Pitting corrosion can be caused by a local break or damage to the protective oxide film or a protective coating; it can also be caused by non-uniformities in the metal structure itself. Pitting is dangerous because it can lead to failure of the structure with a relatively low overall loss of metal.

TYPES OF PITTING CORROSION:



CREVICE CORROSION

Crevice corrosion is also a localized form of corrosion and usually results from a stagnant microenvironment in which there is a difference in the concentration of ions between two areas of a metal. Crevice corrosion occurs in shielded areas such as those under washers, bolt heads, gaskets, etc. where oxygen is restricted. These smaller areas allow for a corrosive agent to enter but do not allow enough circulation within, depleting the oxygen content, which prevents re-passivation. As a stagnant solution builds, pH shifts away from neutral. This growing imbalance between the crevice (microenvironment) and the external surface (bulk environment) contributes to higher rates of corrosion. Crevice corrosion can often occur at lower temperatures than pitting. Proper joint design helps to minimize crevice corrosion.

INTERGRANULAR CORROSION

An examination of the microstructure of a metal reveals the grains that form during solidification of the alloy, as well as the grain boundaries between them. Intergranular corrosion can be caused by impurities present at these grain boundaries or by the depletion or enrichment of an alloying element at the grain boundaries. Intergranular corrosion occurs along or adjacent to these grains, seriously affecting the mechanical properties of the metal while the bulk of the metal remain intact.

An example of intergranular corrosion is carbide precipitation, a chemical reaction that can occur when a metal is subjected to very high temperatures (e.g., 800°F - 1650°F) and/or localized hot work such as welding. In stainless steels, during these reactions, carbon "consumes" the chromium, forming carbides and causing the level of chromium remaining in the alloy to drop below the 11% needed to sustain the spontaneously-forming passive oxide

layer. 304L and 316L are enhanced chemistries of 304 and 316 stainless that contain lower levels of carbon, and would provide the best corrosion resistance to carbide precipitation.

STRESS CORROSION CRACKING (SCC)

Stress corrosion cracking (SCC) is a result of the combination of tensile stress and a corrosive environment, often at elevated temperatures. Stress corrosion may result from external stress such as actual tensile loads on the metal or expansion/contraction due to rapid temperature changes. It may also result from residual stress imparted during the manufacturing process such as from cold forming, welding, machining, grinding, etc. In stress corrosion, the majority of the surface usually remains intact; however, fine cracks appear in the microstructure, making the corrosion hard to detect. The cracks typically have a brittle appearance and form and spread in a direction perpendicular to the location of the stress. Selecting proper materials for a given environment (including temperature and management of external loads) can mitigate the potential for catastrophic failure due to SCC.

GALVANIC CORROSION

Galvanic corrosion is the degradation of one metal near a joint or juncture that occurs when two electrochemically dissimilar metals are in electrical contact in an electrolytic environment; for example, when copper is in contact with steel in a saltwater environment. However, even when these three conditions are satisfied, there are many other factors that affect the potential for, and the amount of, corrosion, such as temperature and surface finish of the metals. Large engineered systems employing many types of metal in their construction, including various fastener types and materials, are susceptible to galvanic corrosion if care is not exercised during the design phase. Choosing metals that are as close together as practicable on the galvanic series helps reduce the risk of galvanic corrosion.

CONCLUSION

In aqueous environments, metals may be exposed to not only uniform corrosion, but also to various types of local corrosion including pitting, crevice, intergranular, stress, and galvanic corrosion. In areas where corrosion is a concern, stainless steel products offer value and protection against these threats. Stainless' favorable chemical composition makes it resistant to many common corrosives while remaining significantly more affordable than specialty alloys such as titanium and Inconel® alloys.

Stainless steel is a highly alloyed, low-carbon steel with a high (at least 11%) chromium content. When exposed to an oxygenated environment, the chromium reacts to form a passive oxide layer on the metal's surface, slowing further oxidation and providing a self-healing quality, which helps resist uniform and local corrosion. Nickel helps to stabilize the microstructure, increasing SCC resistance.

Metal corrosion is a slow process of destruction of metal objects, after chemical reactions that occur on the surface of metals, alloys, under the action of the environment. Corrosion process is influenced by a number of internal and external factors.

Factors affecting rate of corrosion:

Several factors affecting the rate of the corrosion, which can be divided into two parts:

1) Factors affecting on metal (related to the metal)
2) Factors affecting on corrosive environment.

Factors affecting the metals:

1. Nature of the metal: The tendency of the metal to undergo corrosion is mainly dependent on the nature of the metal. IN general the metal with lower electrode potential have more reactive and more susceptible for corrosion and metal with high electrode potential are less reactive and less susceptible for corrosion for egs: metals like K, Na, Mg, Zn etc have low electrode potential are undergo corrosion very easily, where as noble metals like Ag, Au, Pt have higher electrode potential, their corrosion rate are negligible but there are few exception for this general trend as some metals show the property of passivity

like Al, Cr, Ti, Ta etc.

2. Surface state of the metal or nature of the corrosion product (passivity): The corrosion product is usually the oxide of the metal; the nature of the product determines the rate of further corrosion process. If the oxide layer, which forms on the surface, is stoichiometric, highly insoluble and non-porous in nature with low ionic and electronic conductivity then that type of products layer effectively prevents further corrosion, which acts as a protective film. For egs: Al, Cr, Ti develop such a layer on their surface and become passive to corrosion and some metal like Ta, Zr and Mo not only forms such a protective layers but are capable of self repairing oxide films when it is damaged. Hence these are extremely passive metals.
If the oxide layer forms on the metal surface is non-stochiometric, soluble, unstable and porous in nature and have an appreciable conductivity, they cannot control corrosion on the metal surface for egs: oxide layer formed on metals like Zn, Fe, Mg etc.

3. Anodic and Cathodic area:

The rate of the corrosion is greatly influenced by the relative sizes of cathodic and anodic areas. If the metal has smaller the anodic area and larger the cathodic area exposed to corrosive atmosphere, more intense and faster is the corrosion occurring at anodic area because at anode oxidation takes place and electrons are liberated. At the cathode these electrons are consumed. When anode is smaller and cathode region is larger all the liberated

electrons at anode are rapidly consumed. This process makes the anodic reaction to takes place at its maximum rate thus increasing the corrosion rate. If the cathode is smaller and reverse process takes place decrease rate of corrosion.

For egs: If tin (Sn) coated on iron (Fe) and in that some area are not covered or some pin holes are left, there forms smaller anodic area and larger cathodic area because tin is cathodic with respect to iron so intense localized corrosion takes place. On the other hand if Zn coated to Fe then if there are some pin holes are there creates larger anodic area and smaller cathodic area because Fe is cathodic with respect to zinc so that rate of corrosion is veryless.

4. Hydrogen over voltage:

A metal with low hydrogen over voltage on its surface is more susceptible for corrosion. When the cathodic reaction is hydrogen evolution type with low hydrogen over voltage, liberation of H2 gas is more easier so that cathodic reaction is very fast, that makes anodic reaction faster hence overall corrosion process is very fast. If the H2 over voltage is high so cathodic reaction is slow hence corrosion reaction also slower.

Corrosion Protection & Prevention:

In virtually all situations, metal corrosion canbemanaged,slowed, or even stopped by using the proper techniques. Corrosion prevention can take a number of forms depending on the circumstances of the metal being corroded. Corrosion prevention techniques can be generally classified into 6 groups:

1) **Environmental Modification**

Corrosion is caused by chemical interactions between metal and gasses in the surrounding environment. By removing the metal from, or changing, the type of environment, metal deterioration can be immediately reduced.

This may be as simple as limiting contact with rain or seawater by storing metal materials indoors or could be in the form of direct manipulation of the environment affecting the metal.

Methods to reduce the sulfur, chloride, or oxygen content in the surrounding environment can limit the speed of metal corrosion. For example, feed water for water boilers can be treated with softeners or other chemical media to adjust the hardness, alkalinity or oxygen content in order to reduce corrosion on the interior of the unit.

2)Metal Selection and Surface Conditions

No metal is immune to corrosion in all environments, but through monitoring and understanding the environmental conditions that are the cause of corrosion, changes to the type of metal being used can also lead to significant reductions in corrosion.

Metal corrosion resistance data can be used in combination with information on the environmental conditions to make decisions regarding the suitability of each metal.

The development of new alloys, designed to protect against corrosion in specific environments, is constantly under production. Hastelloy nickel alloys, Nirosta steels, and Timetal titanium alloys are all examples of alloys designed for corrosion prevention.

Monitoring of surface conditions is also critical in protecting against metal deterioration from corrosion. Cracks, crevices or asperous surfaces, whether a result of operational requirements, wear and tear, or manufacturing flaws, all can result in greater rates of corrosion.

Proper monitoring and the elimination of unnecessarily vulnerable surface conditions, along with taking steps to ensure that systems are designed to avoid reactive metal combinations and that corrosive agents are not used in the cleaning or maintenance of metal parts are all also part of effective corrosion reduction program.

3)Cathodic Protection

Galvanic corrosion occurs when two different metals are situated together in a corrosive electrolyte.

This a common problem for metals submerged together in seawater, but can also occur when two dissimilar metals are immersed in close proximity in moist soils. For these reasons, galvanic corrosion often attacks ship hulls, offshore rigs, and oil and gas pipelines.

Cathodic protection works by converting unwanted anodic (active) sites on a metal's surface to cathodic (passive) sites through the application of an opposing current. This opposing current supplies free electrons and force local anodes to be polarized to the potential of the local cathodes.

Cathodic protection can take two forms. The first is the introduction of galvanic anodes. This method, known as a sacrificial system, uses metal anodes, introduced to the electrolytic environment, to sacrifice themselves (corrode) in order to protect the cathode.

While the metal needing protection can vary, sacrificial anodes are generally made of zinc, aluminum, or magnesium, metals that have the most negative electro-potential. The galvanic series provides a comparison of the different electro-potential - or nobility - of metals and alloys.

In a sacrificial system, metallic ions move from the anode to the cathode, which leads the anode to corrode more quickly than it otherwise would. As a result, the anode must regularly be replaced.

The second method of cathodic protection is referred to as impressed current protection. This method, which is often used to protect buried pipelines and ship hulls, requires an alternative source of direct electrical current to be supplied to the electrolyte.

The negative terminal of the current source is connected to the metal, while the positive terminal is attached to an auxiliary anode, which is added to complete the electrical circuit.

Unlike a galvanic (sacrificial) anode system, in an impressed current protection system, the auxiliary anode is not sacrificed.

4)Inhibitors

Corrosion inhibitors are chemicals that react with the metal's surface or the environmental gasses causing corrosion, thereby, interrupting the chemical reaction that causes corrosion.

Inhibitors can work by adsorbing themselves on the metal's surface and forming a protective film. These chemicals can be applied as a solution or as a protective coating via dispersion techniques.

The inhibitor's process of slowing corrosion depends upon:

- Changing the anodic or cathodic polarization behavior
- Decreasing the diffusion of ions to the metal's surface
- Increasing the electrical resistance of the metal's surface

Major end-use industries for corrosion inhibitors are petroleum refining, oil and gas exploration, chemical production and water treatment facilities. The benefit of corrosion inhibitors is that they can be applied in-situ to metals as a corrective action to counter unexpected corrosion.

5)Coatings

Paints and other organic coatings are used to protect metals from the degradative effect of environmental gasses. Coatings are grouped by the type of polymer employed. Common organic coatings include:

- Alkyd and epoxy ester coatings that, when air dried, promote cross-link oxidation
- Two-part urethane coatings
- Both acrylic and epoxy polymer radiation curable coatings
- Vinyl, acrylic or styrene polymer combination latex coatings
- Water-soluble coatings
- High-solid coatings
- Powder coatings

6)Plating

Metallic coatings, or plating, can be applied to inhibit corrosion as well as provide aesthetic, decorative finishes. There are four common types of metallic coatings but the common one is given below.

- Electroplating: A thin layer of metal - often nickel, tin, or chromium - is deposited on the substrate metal (generally steel) in an electrolytic bath. The electrolyte usually consists of a water solution containing salts of the metal to be deposited.

Questions:

1. Which series (Galvanic/ Electrochemical) predicts the relative corrosive tendencies of a metal?
2. Galvanic corrosion is also differential metal corrosion.
3. Out of copper, chromium and iron which is unstable and protective?
4. State the reason for : wire mesh corrodes faster at the joints./ corrosion occurs in steel pipe connected to copper plumbing
5. How to prevent corrosion of metals due to moisture?
6. Give basic difference between galvanization and tinning.
7. soil having more porosity will have high corrosion.
8. Hygroscopic suspended particles cause more corrosion.
9. Annealing is a useful process.
10. Smaller the grain size of metal, faster is corrosion.
11. SO_2 and H_2S increase corrosion.
12. L,T and V type joints should be avoided.
13. Sn undergoes severe corrosion in presence of chlorine but not Ag.
14. Sn-coated iron articles are free from any damage.
15. Part of nail inside the wood undergoes corrosion.
16. Bolts and nuts may be made of same metals.
17. Deposition of small dust particles on metal surface must be avoided.
18. Mg sheets are best suited for protection.
19. Brass containers are not cleaned by alkaline solution.
20. Justify that corrosion is a diffusion process.

Module – 4: ENERGY SCIENCE

A fuel is any combustible material that can be made to react with other substances so that it releases energy.

Types of Fuels: Fuels can be generally classified into two types.

1. *On the basis of their fuels state/ physical state of aggregation:*
    - Solid Fuels
    - Liquid Fuels
    - Gaseous Fuels
2. *On the basis of their occurrence/ source:*
    - Natural Fuels
    - Artificial Fuels

Examples of Fuels:

Natural Fuels	Artificial Fuels
Solid Fuels	
Wood, Coal, Oil Shale	Tanbark, Bagasse, Straw, Charcoal, Coke Coal, Briquettes
Liquid Fuels	
Petroleum	Oils from distillation of petroleum, Coal Tar, LPG, Shale-Oil, Alcohols, etc.
Gaseous Fuels	
Natural Gas	Coal gas, Bio gas, CNG Producer Gas, Water Gas, Hydrogen, Acetylene, Blast Furnace Gas, Oil Gas

Solid Fuels

Advantages:

- Easier transportation and storage.
- Low production cost.
- Moderate ignition temperature.

Disadvantages:

- Large portion of energy is wasted.
- Cost of handling is high and controlling is also hard.
- Ash content is high & burn with clinker formation.

Liquid Fuels:

Advantages:

- Higher calorific value per unit mass.
- Burn without ash, clinkers, etc.
- Controlling the combustion is easier.
- Transportation easier through pipes and stored indefinitely without loss.
- Loss of energy is comparatively lower.
- Require less furnace space for combustion.

Disadvantages:

- Cost of liquid fuel is much higher compared to solid fuel.
- Storage methods are costlier.
- Greater risk of fire hazards.
- Special burning equipment required for more efficient combustion.

Gaseous Fuels:

Advantages:

- Transportation through pipes is easy.
- Sparking combustion is really easy.
- They have a higher heat content.

- Clean after use.
- Do not require any special burner technology.

Disadvantages:

- Large storage tanks required.
- As they are highly inflammable, the chance for fire hazards are extremely high and strict safety measures need to be followed.

Fossil Fuel:

Fossil fuels are the dead and decayed remains of plants and animals subjected to decades of pressure and temperature under the earth's crust. Primarily fossil fuels are hydrocarbons. They are convenient and effective. They provide the calorific value required to fulfill our needs. Even though they are available in plenty right now, they are a non-renewable source of energy. The burning of fossil fuels is responsible for a large section of the world's pollution index.

Types of fossil fuels:

- *Coal*
- *Oil*
- *Natural Gas*

Characteristics of a good fuel

1) High calorific value 2) Moderate ignition temp as low temp can cause fire hazard where as high value makes it difficult to ignite.3) low moisture content 4) low non-combustible matter. Other factors include moderate rate of combustion, low cost, uniform size.

calorific value (plural calorific values):

The amount of energy produced by the complete combustion of a material or **fuel**. Measured in units of energy per amount of material, e.g. kJ/kg. The amount of energy available from an item of food when digested, mostly from carbohydrates and fats. The calorific value is a characteristic for each substance. It is measured in units of **energy** per unit of the substance, usually **mass**, such as: kcal/**kg**, kJ/**kg**, J/mol, Btu/m^3. Heating value is commonly determined by use of a bomb calorimeter.

1 calorie--- Heat required to increase temp. of 1gm of water through 1^0c.

1 B.Th. U---- Amount of heat reqd. to raise temp of 1 pound of water through 1^0F.

Centrigrade Heat Unit----Quanty of heat required to raise temp. of 1 pound of water through 1^0c.

1 Kcal =39.68 B.Th.U =2.2C.H.U

HCV:

HCV is the amount of heat or energy liberated from the combustion of a fuel when its byproducts are condensed to recover some of the energy released from the combustion. In comparison, **LCV** is the amount of energy liberated from the combustion of a fuel when its byproducts freely escape. During the use of any fuel, water vapors and moisture etc. escape as such along with the combustion gases. As they are not condensed, a lower amount of heat is available.

LCV = HCV- Latent heat of vapors formed

Since 1 part by mass of hydrogen produces 9 parts by mass of water.

Hence, LCV= HCV — mass of H_2 × Latent heat of steam

Latent heat of steam = 587Kcal/Kg or 1060 B.T.U/lb

Unit Conversion: 1Kcal/Kg = 1.8 B. T .U

1Kcal/ m3 = 0.1077 B.T.U/ft^3

1B.T.U/ft3 = 9.3 K cal/ m^3

<u>Calculation</u> If oxygen is present in the fuel, it is in the combination with H2 as fixed hydrogen (H2O). 2H2 +O2 → 2H2O

8 parts of O2combine with1 part of H2.

Fixed hydrogen = mass of oxygen in fuel/ 8

Amount of H2 available for combustion =Total mass of H2 in the fuel – fixed H2 =(H-O/8)

H2O formed from 1 gm of H2 = 9 gm

H2O formed from H/100 gm of H2 = 9X H/100 gm = 0.09H gm

Latent Heat of steam = 587 cal/gm, Latent heat of water vapour = 0.09H X 587 cal/gm.

LCV = HCV-0.09X587 cal/gm

Theoretical Calculation of calorific value of a fuel:

Calorific value of a fuel is the sum of the calorific value of all components present.

Constituent	Hydrogen	Carbon	Sulphur
HCV(Kcal/Kg)	34,500	8080	2420

Dulong's Formula : HCV = **1/100**[8080C +34500(H-O/8) +2240S] Kcal/kg

Calorific value of Fuels:

Wood - 2500 kcal/kg

CNG-- 4900 kcal/m3

Producer gas- 2800 kcal/m3

Natural gas---12000 kcal/kg

LPG—27800 kcal/kg

Kerosene—11100 kcal/kg

Diesel- 11000 kcal/kg

Petrol—11250 kcal/kg

Electricity--- 860 kcal/kwh

Fractional distillation of crude oil:

Fractional distillation separates a mixture into a number of different parts, called **fractions**.

A tall fractionating column is fitted above the mixture, with several condensers coming off at different heights. The column is hot at the bottom and cool at the top. Substances with high boiling points condense at the bottom and substances with lower boiling points condense on the way to the top.

Crude oil is a mixture of hydrocarbons. The crude oil is evaporated and its vapours condense at different temperatures in the fractionating column. Each fraction contains hydrocarbon molecules with a similar number of carbon atoms and a similar range of boiling points.

Oil fractions:

The diagram below summarizes the main fractions from crude oil and their uses, and the trends in properties. Note that the **gases leave at the top of the column**, the **liquids condense in the middle** and the **solids stay at the bottom**.

The fractionating column
As we go up the fractionating column, the hydrocarbons have:

1. lower boiling points
2. lower viscosity (they flow more easily)
3. higher flammability (they ignite more easily).

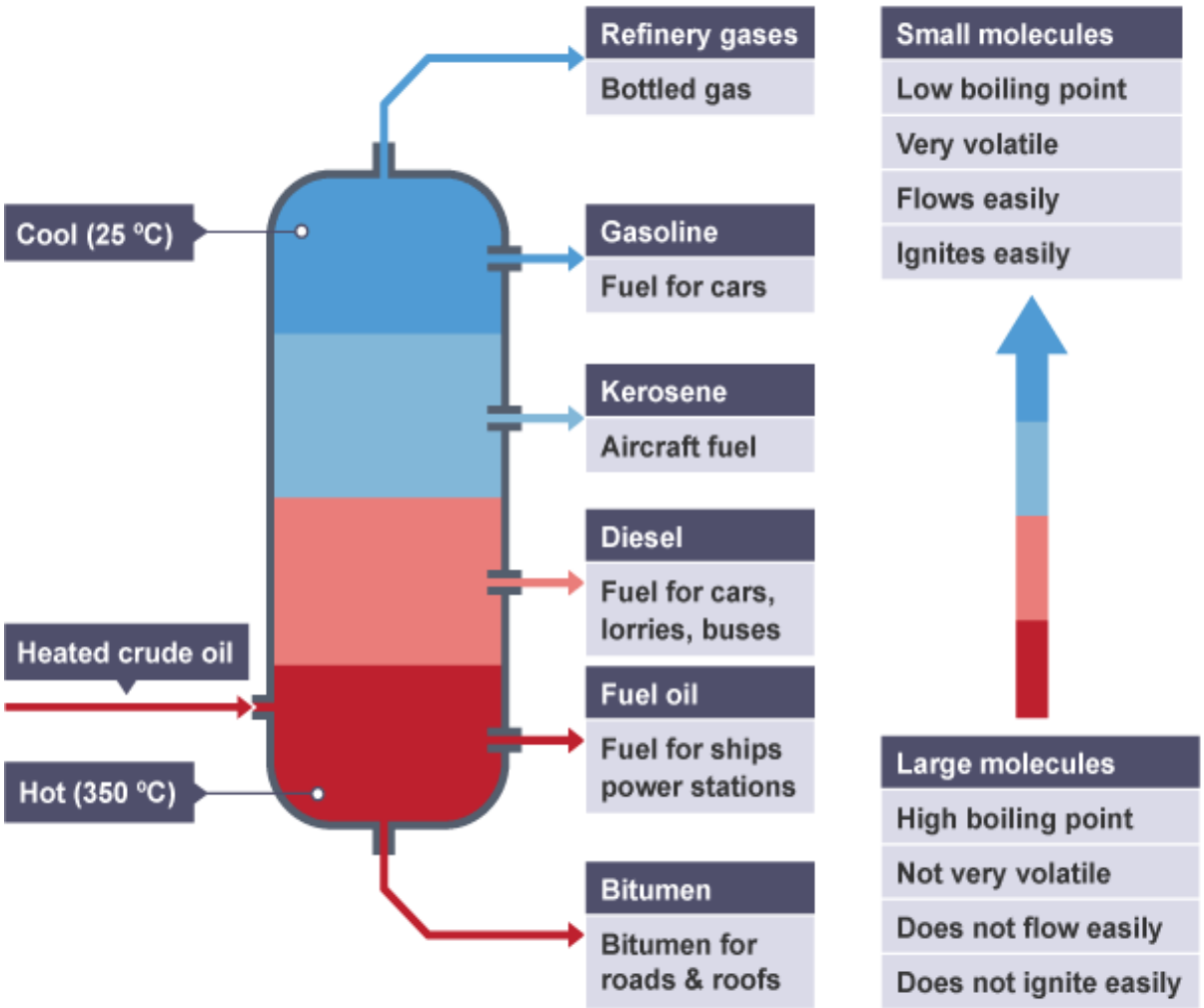
The various components of crude oil have different sizes, weights and boiling temperatures; so, the first step is to separate these components. Because they have different boiling temperatures, they can be separated easily by a process called **fractional distillation**. The steps of fractional distillation are as follows:

1. You **heat** the mixture of two or more substances (liquids) with different boiling points to a high temperature. Heating is usually done with high pressure steam to temperatures of about 1112 degrees Fahrenheit / 600 degrees Celsius.
2. The mixture **boils**, forming vapor (gases); most substances go into the vapor phase.
3. The **vapor** enters the bottom of a long column (**fractional distillation column**) that is filled with trays or plates. The trays have many holes or bubble caps (like a loosened cap on a soda bottle) in them to allow the vapor to pass through. They increase the contact time between the vapor and the liquids in the column and help to collect liquids that form at various heights in the column. There is a temperature difference across the column (hot at the bottom, cool at the top).
4. The **vapor rises** in the column.
5. As the vapor rises through the trays in the column, it **cools**.
6. When a substance in the vapor reaches a height where the temperature of the column is equal to that substance's boiling point, it will **condense** to form a liquid. (The substance with the lowest boiling point will condense at the highest point in the column; substances with higher boiling points will condense lower in the column.).
7. The trays **collect** the various liquid fractions.
8. The collected liquid fractions may pass to condensers, which cool them further, and then go to storage tanks, or they may go to other areas for further chemical processing

Fractional distillation is useful for separating a mixture of substances with narrow differences in boiling points, and is the most important step in the refining process.

Before fractional distillation, water is separated by Cottrell's process which allows the oil to flow between two highly charged **electrodes**. The colloidal water-droplets coalesce to form a large drop, which separate out. Similarly, harmful 'S' compounds can be removed by treatment with copper oxide to give solid copper sulphide that is filtered out.

Fraction distilled from crude oil	Boiling point range (°C)	Carbon chain length	Hydrocarbons present	Uses	Viscosity	Flammability
Refinery Gas	-160 to -5	1-4	Methane CH_4 Ethane C_2H_6 Propane C_3H_8 Butane C_4H_{10}	Home heating and cooking, camping fuel	Low -easily flows (*as gas or runny liquid*)	High (ignites and burns easily)
Gasoline(petrol)	40-110	5-8	Octane C_8H_{18}	Car fuel	viscosity increases	flammability decreases
Naphtha	110-180	8-10	Decane $C_{10}H_{22}$	Plastics		
Kerosene (paraffin)	180-260	10-16	Dodecane $C_{12}H_{26}$	Jet aircraft fuel		
Diesel	260-320	16-20	Hexadecane $C_{16}H_{34}$	Fuel for buses and lorries		
Fuel Oil	320-400	20-50	Icosane $C_{20}H_{42}$	Industrial heating systems	High-does not flow easily (*thick sticky consistency*)	Low (does not ignite or burn easily, burns with sooty flame)
Bitumen/ Residue	400-600	>50		Surfacing roads		



Diesel

Cheaper, less consumption per unit power produced, Thermal efficiency higher, Exhaust gases contain less pollutants(CO_2, Hydrocarbon), Expensive fuel injection device

Petrol costilier, more consumption., lower thermal efficiency, Emit higher pollutants, simple IC engine

Refining of gasoline: Very much necessary as it contains some undesirable unsaturated hydrocarbons and sulphur compounds. The former gets oxidized & polymerized, thus causing gums & sludge formation on storing. Sulphur compounds corrode IC engines.

Process: Sulphur compounds are generally removed by treating with alkaline solution of sodium plumbite with controlled addition of sulphur. This process is called 'Sweetening of petrol' .Final conversion is known as 'Doctor's Process'.

$2RSH + Na_2Pbo_2 \rightarrow Pb(SR)_2 + 2NaOH$

$Pb(SR)_2 \rightarrow PbS_s + RSSR$

The disulphides are extracted with a suitable solvent.

Olefins & colouring matters of gasoline are removed by percolating gasoline through Fuller's earth , which adsorbs the other olefins & colours. After refining, some inhibitors are added. These retard oxidation process thereby improving the storing.

Cracking of petrol:

Cracking, in **petroleum** refining, is the process by which heavy hydrocarbon molecules are broken up into lighter molecules by means of heat and usually pressure and sometimes catalysts. **Cracking** is the most important process for the commercial production of gasoline and diesel fuel.

$\mathbf{C_{10}H_{22} \rightarrow C_5H_{12} + C_5H_{10}}$

Decane n-pentane pentene

Types of Cracking

Thermal **Cracking** and Catalytic **Cracking**. **Cracking** is a process by which complex high molecular weight organic compounds are broken down into smaller fragments of molecules. The complex high molecular weight organic compounds are generally long chained hydrocarbons such as petroleum. Thermal is further divided in to **liquid & vapour phase. Catalytic is of fixed bed & moving bed.**

Thermal cracking: No catalyst used heavy oils are subjected to high temp & pressure.

Characteristics	Liquid Phase	Vapour phase
Cracking temp	475-5300c	600-630
Pressure	100 kg/cm2	10-20 KG/ cm2
Yield of petrol	50-60 %	> 60%
Octane rating	65-70	> 70
Time reqd.	More	less

Catalytic Cracking: Much lower temp& pressure than thermal/ Suitable catalyst used are Al_2O_3, $Al_2(SiO_3)_3$, Better yield & quality, a low pressure reqd / no undesired product as majority escapes as H_2S / product with high octane rating/ use of selective catalyst/ no external fuel needed as fuel comes from coal embedded in catalyst.

Knocking:

In an internal-combustion engine, sharp sounds caused by premature combustion of part of the compressed air-fuel mixture in the cylinder. In a properly functioning engine, the charge burns with the flame front progressing smoothly from the point of ignition across the combustion chamber. However, at high compression ratios, depending on the composition of the fuel, some of the charge may spontaneously ignite ahead of the flame front and burn in an uncontrolled manner, producing intense high-frequency pressure waves. These pressure waves force parts of the engine to vibrate, which produces an audible knock. Compression ratio (CR) is the ratio of gaseous volume(V1) in the cylinder at the end of the suction-stroke to the volume(V2) at the end of compression –stroke of the piston.

As V2 < V1, CR = V1/V2 > 1. The 'CR' at which the fuel tends to knock is known as '*critical compression ratio'.*

Consequences:

Knocking can cause overheating of the spark-plug points, erosion of the combustion chamber surface, and rough, inefficient operation. It can be avoided by adjusting certain variables of engine design and operation, such as compression ratio and burning time; but the most common method is to burn gasoline of higher octane number. Higher the **octane** number, the more compression from the piston the gasoline can withstand before igniting in the engine. The ability for gasoline to withstand being compressed is most **important** for vehicles that run on traditional gasoline.

The octane number of gasoline is a measure of its resistance to knock. The octane number is determined by comparing the characteristics of a gasoline to isooctane (2,2,4-

trimethylpentane) and heptane. Isooctane is assigned an octane number of 100. It is a highly branched compound that burns smoothly, with little knock. Octanerating canbe improved with octane boosters such as ethanol, di ethyl Telluride ,methanol, MTBE, ETBE (all oxygenated and somewhat safe) or other chemicals. In the past it was usually TEL (lead based, which were banned in the mid-70's since they're toxic and also destroyed the car's catalytic converters).

Tetra ethyl Lead: **$(C_2H_5)_2pb$** forms Pb and PbO particles which react as chain reaction inhibitors by slowing down the chain oxidation reaction responsible for early detonation leading to knocking sound.

1)Presence of 'S' reduces the efficiency of TEL . So, desulphurised Gasoline is used.

2) Pb & PbO_2 decrease engine life so these are removed as follows, $Pb + C_2H_4Br_2 \rightarrow PbBr_2 + C_2H_4$, $PbBr_2$ is volatile so escapes along with exhaust gas.

3) Another cause of pollution is incomplete combustion leading to the formation of CO_X, NO_X etc. Hence catalytic convertors based on Pt are employed to carry out catalytic combustion reaction. But lead present in exhaust gas poisons active sites in the catalyst. Catalytic converter is a Pt or Rh based compound which helps to convert more poisonous CO to CO_2.

$CO \rightarrow CO_2$. But Pt is poisoned by Pb, hence the need for unleaded petrol.

MTBE : It is preferred as it contains oxygen in the form of ether group and supplies oxygen for combustion there by reducing the formation of peroxy compound. Its advantage is that it permits use of catalytic converter.

Octane numbers:

- The octane number of a fuel is based on how fuels auto ignite, under compression.
- It is based on the scale where isooctane is 100 (minimal ignition under pressure) and heptane is 0 (easily ignites under pressure).
- For example, a petrol with an octane number of 92 has the same knock as a mixture of 92% isooctane and 8% heptane.
- Octane rating decreases with an increase in the carbon chain length.
- Octane ratings increase with carbon chain branching.
- Octane ratings increase in aromatics with same number of carbons.
- It is important to know octane numbers for petrol because the auto ignition of fuels causes a knocking effect in petrol engines. This is where it ignites twice; once due to the high pressure and again when the spark ignites the petrol. This causes the car engine to be less efficient and it can also damage the engine.

- However, diesel engines rely on this knocking effect, as they have no spark plugs and rely on the effect of compression to make their fuel/air mixtures ignite.
- **<u>Octane rating order</u> is given in decreasing order as:**

  n-alkanes > mono substituted alkanes> cyclic alkanes > alkenes > alkenes > poly substituted alkenes > aromatics.

  Similarly for straight chain hydrocarbons, **n-hexane > n-pentane > n-butane**

  'Octane Rating' was introduced by Graham Edger in 1872 which gave n-heptane with zero with iso octane 100. The best gasoline known as 80-octane fuel is having 80:20 combination of iso octane & n-heptane.

How octane rating of a fuel can be increased?

- There are two ways to increase the octane number of a fuel. One is to put special additives into the fuel which discourage auto ignition, and the other is to blend high-octane fuels in with the ordinary petrol.

The right kind of alkane:

- The best kind of alkane for fuels is not the most plentiful in crude oil.
- The shorter the alkane chain, the higher the octane number. However, the volatility of the hydrocarbon increases with a decrease in the chain length.
- The short alkane chains have to be mixed in petrol to lower their volatility and higher the octane number of the petrol.
- The other factor that affects the octane number is the degree of branching in the alkane chain. The more branched the alkane is, the higher the octane number.
- Crude oil contains both straight-chain and branched alkanes, but it does not contain enough branched alkanes to give it a high octane number.
- But this can be solved by increasing the octane numbers by isomerism, reforming and cracking.

Isomerism

- Isomerization involves heating up straight chained alkanes in the presence of suitable catalysts, so that the chains break.
- When the chains join together again, they are more likely to be branched.

- Oil refineries do this on a large scale with C_6H_{14} and C_5H_{12}.
- The isomerization reaction is reversible and higher temperatures favour the production of straight chained isomers. However the temperature cannot be too low or the reaction would take too long, so a compromise is made between yield and rate of reaction. The temperature used is 250°C.

Zeolite

- The catalyst involved in the isomerization is platinum; the products are then passed over a form of zeolite, which acts as a molecular sieve, separating the straight chains from the branched chains.

Cetane Number OR Rating:

It is a measure of the ignition value of a diesel fuel that represents the percentage by volume of cetane(Hexadecane,$C_{16}H_{34}$) in a mixture of liquid methylnaphthalene that gives the same ignition lag as the oil being tested.

Cetane is a colorless, liquid hydrocarbon (a molecule from the alkane series) that ignites easily under compression. For this reason, it was given a base rating of 100 and is used as the standard measure of the performance of compression ignition fuels, such as diesel and biodiesel. All the sundry hydrocarbon constituents of diesel fuel are measured and indexed against cetane's base-100 rating.

Cetane rating is given as: **n-alkane > napthalenes > alkenes > branched alkanes > aromatics**

From Left to Right: Ignition delay increases**,** Ignition quality decreases, Cetane number decreases

Cetane Number (Significance)

Similar to the octane number rating that's applied to gasoline to rate its ignition stability, the cetane number is the rating assigned to diesel fuel to rate its combustion quality. While gasoline's octane number signifies its ability to resist auto-ignition (also referred to as pre-ignition, knocking, pinging, or detonation), diesel's cetane number is a measure of the fuel's delay of ignition time (the amount of time between the injection of fuel into the combustion chamber and the actual start of combustion of the fuel charge).

Because diesel engines rely on compression ignition (no spark), the fuel must be able to auto-ignite—and generally, the quicker the better. A higher cetane number means a shorter ignition delay time and more complete combustion of the fuel charge in the combustion chamber. This, of course, translates into a smoother running, better performing engine. Cetane number requirements depend mainly on engine design, size , speed of operation, and

load variations—and to a slightly lesser extent, atmospheric conditions. Conversely, running a diesel engine on fuel with a lower-than-recommended cetane number can result in rough operation (noise and vibration), low power output, excessive deposits and wear, and hard starting.

Cetane Numbers of Various Diesel Fuels:

Normal modern highway diesel engines run best with a fuel rated between 45 and 55. The following list of cetane numbers denotes varying grades and types of compression ignition diesel fuels:

- Regular diesel: 48
- Premium diesel: 55
- Biodiesel (B100): 55
- Biodiesel blend (B20): 50
- Synthetic diesel: 55
- Commonly High speed diesel(45-60), Low speed(~25).

Additives to raise cetane number : Alkyl nitrates (principally 2-ethylhexyl nitrate) and di-tert-butyl peroxide , acetone peroxide are used as additives to raise the cetane number.

Liquid Fuels:

LPG: It is obtained as a by product during cracking of heavy oils or natural gas. It is dehydrated, desulphurised & traces of odorous organic sulphides (Mercaptans) is added to give warnings of gas leak. LPG is supplied under pressure in containers under various trade name. Its calorific value is 27,800 Kcal/m^3. It consists of hydrocarbons of such volatility so that it can exist as gas under atmospheric pressure but can also be easily liquefied under pressure. Mainly consists of n-butane,, isobutene, butylenes and propane with little or no amount of ethane and propylene.

Advantages over Gasoline: High knock-resistant, cheaper, burns cleanly where as disadvantages include its handling only under pressure with not being easily detectable.

Non-Petroleum Fuels: Mostly used as an alternative fuel.

Power Alcohol : Fuel containing 25% ethanol.

$C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6 + C_6H_{12}O_6$ (In presence of Invertase)

$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$(In presence of Zymase)

The above process takes 36 to 38 hours for complete fermentation. The fermented liquid(WASH) contains about 18-20% alcohol. The wash is then fractionally distilled to get

"Rectified Spirit"(90-95% alcohol). To get absolute alcohol(100 %), the rectified spirit is digested with lime for 2 days and distilled.

Natural Gases: It is called '**Wet Gas'** and it is processed to remove propane, propene, butane and butane. When the gas is associated with crude oil, it is called '**dry gas'**. Approximate component of natural gas: CH_4 = 70%, C_2H_6 = 5-10%, H_2 =3%,($CO + CO_2$) = Rest, calorific value = 12000-14000 kcal/ m^3 or (12500 kcal/kg.)

CNG : A natural gas compressed to a high pressure of about 1000 atm Say a steel cylinder containing 15 kg of CNG contains $2x10^4$ or $20m^3$of natural gas at 1 atm. pressure**.** It is a good substitute for petrol or diesel as it gives no sulphur or nitrogen during combustion and no carbon particles.

Coal Gas: When coal is carbonized or heated in the absence of air at about 1300^0c, we get coal gas.

Coal → coke + coal gas

Properties: It is a colourless gas having a characteristic odour. It is lighter than air and burns with a long smoky flame. Avg. composition: H_2 =40%, CH_4 = 32%, CO =7%, C_2H_2=2%, C_2H_4 =4%, N_2 =4%, CO_2=1% and rest =4%. Calorific value = 4,900Kcal/ m^3

Uses; As a fuel/ In metallurgical operations for providing reducing atmosphere.

Oil Gas : Prepared by cracking of Kerosene oil in a coal-fixed furnace.

$C_{12}H_{26}$ → CH_4+ C_2H_6+C_2H_2+ C_2H_4+ C_3H_8

Calorific value is 4500-5400 Kcal/ m^3. It is used as a laboratory gas with the composition, CH_4 =25-30%, H_2=50-55%, CO= 10-20%, CO_2 = 3%

Producer Gas: A mixture of combustible gases (CO + NO2) associated with a large amount of non-combustible gases like N2, CO2.

It is prepared bypassing air mixed with a little of steam over a red hot coal bed maintained at about 10000c in a special reactor, called 'Gas

Producer'. Calorific value =1300 Kcal/m^3, Avg. composition ; CO =22.3%, H_2 =8.12%, N_2= 52%, CO_2 =3%.

Uses: Useful for heating in a furnace, muffle furnace, , retorts or as a reducing agent in metallurgical operation.

<u>Water Gas</u>: A mixture of combustible gases(CO2+ H2) with a little amount of non-combustible gases like CO2 and N2. It is produced by passing alternatively steam and little air through a bed of red hot coal/ coke maintained at about 900-10000C in a reactor.

$C + H_2O \rightarrow CO + H_2$ -29Kcal/mole

Calorific Value = 2800 Kcal/m^3

Avg. composition : H_2= 51%, CO=41%, N_2 =4%, CO_2 = 4%

Uses; A source of H2, an illuminating gas, as a fuel also known as 'Blue Gas'

Synthetic Fuel: A liquid or a gaseous fuel obtained from syn gas (a mixture of CO & H2) where as Syn gas is derived from gasification of coal gas or biomass.

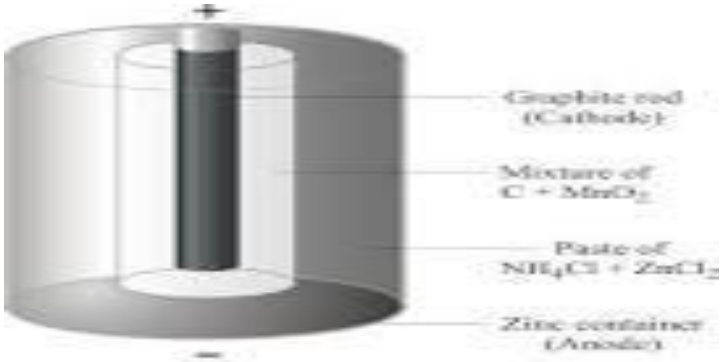
BATTERY SCIENCE:

A battery is a device consisting of one or more electrochemical cells with external connections for powering electrical devices such as flashlights, mobile phones, and electric cars. When a battery is supplying electric power, its positive terminal is the cathode and its negative terminal is the anode. The terminal marked negative is the source of electrons that will flow through an external electric circuit to the positive terminal. When a battery is connected to an external electric load, a redox reaction converts high-energy reactants to lower-energy products, and the free-energy difference is delivered to the external circuit as electrical energy.

Primary (single-use or "disposable") batteries are used once and discarded, as the electrode materials are irreversibly changed during discharge; a common example is the alkaline battery used for flashlights and a multitude of portable electronic devices. Secondary (rechargeable) batteries can discharge and recharged multiple times using an applied electric current; the original composition of the electrodes can be restored by reverse current. Examples include the lead-acid batteries used in vehicles and lithium-ion batteries used for portable electronics such as laptops and mobile phones.

DRY CELL:

Dry cell or Leclanche cell is a primary cell having a zinc anode, a carbon (graphite) cathode surrounded by manganese dioxide, and a paste containing ammonium chloride as electrolyte. The electromotive force (emf) produced by a dry cell is 1.5 V. Dry cell is not reversible and therefore have a limited life time. Dry cell batteries create electrical energy by converting chemical energy into electricity. These materials are placed within the electrolyte paste within the battery. They react with each other through a chemical process in which the electrolyte (carbon or manganese dioxide) reacts with the zinc, creating electricity perating life.



Advantages of Leclanche' Battery

1. The cost of this battery cell is quite low.
2. Various shapes, sizes and capacities of these cells are easily available.
3. Long traditional reliability.

Disadvantages of Leclanche' Battery

1. Its energy density is quite low.
2. It gives poor service in low temperature.
3. It has poor leakage resistance.
4. Cannot perform efficiently at high current drain application.
5. Self-life is not very good.
6. Its voltage falls steadily with discharge.

The overall reaction occurring in the dry cell produces **1.5 volts**:

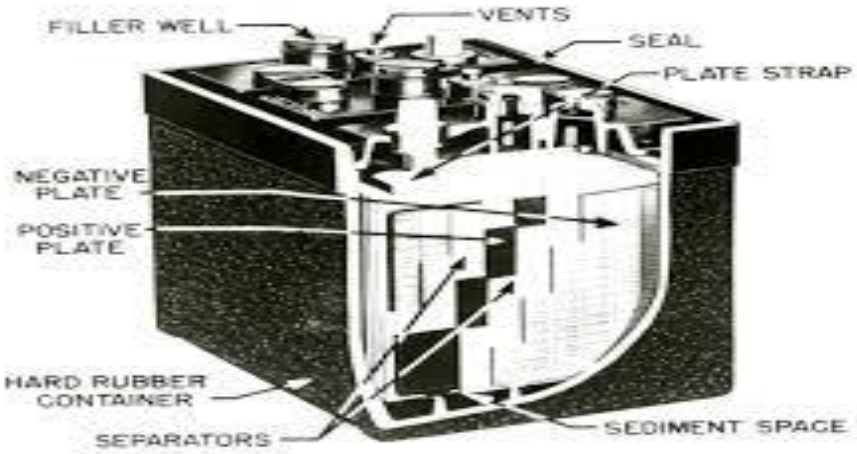
anode			**$Zn(s)$**	$\longrightarrow$	**$Zn^{2+}(aq)$**	**+**	**$2e^-$**
cathode	**$2\ e^-$**	**+**	**$2NH_4^+(aq)$**	$\longrightarrow$	**$2\ NH_3(g)$**	**+**	**$H_2(g)$**
	$2\ NH_3(g)$	**+**	**$Zn^{+2}\ (aq)$**	$\longrightarrow$	**$[Zn(NH_3)_2]^{2+}(aq)$**		
	$H_2(g)$	**+**	**$2\ MnO_2(s)$**	$\longrightarrow$	**$Mn_2O_3(s)$**	**+**	**$H_2O(l)$**

$$Zn(s) + 2NH_4^+(aq) + 2\ MnO_2(s) \longrightarrow [Zn(NH_3)_2]^{2+}(aq) + Mn_2O_3(s) + H_2O(l)$$

WET CELL BATTERY:

There are huge chemical process is involved in Lead Acid battery's charging and discharging condition. The diluted sulfuric acid H2SO4 molecules break into two parts when the acid dissolves. It will create positive ions 2H+ and negative ions SO4-. As we told before, two electrodes are connected as plates, Anode and Cathode. Anode catches the negative ions and cathode attracts the positive ions. This bonding in Anode and SO4- and Cathode with 2H+ interchange electrons and which is further react with the H2O or with the water (Diluted sulfuric acid, Sulfuric Acid + Water).

The battery has two states of chemical reaction, **Charging and Discharging**.



Discharging of a lead acid battery is again involved with chemical reactions. The sulfuric acid is in the diluted form with typically 3:1 ratio with water and sulfuric acid. When the loads are connected across the plates, the sulfuric acid again breaks into positive ions 2H+ and negative ions SO4. The hydrogen ions react with the PbO2 and make PbO and water H2O. PbO start reacting with the H2SO4 and creates PbSO4 and H2O.

On the other side SO4- ions exchange electrons from Pb, creating radical SO4 which further creates PbSO4 reacting with the Pb.

As explained above, following chemical reactions takes place at Anode and Cathode during the discharging process. These reactions are exactly opposite of charging reactions:

At cathode

$$\mathbf{Pb + SO_4^{2-} \Rightarrow PbSO_4 + 2e^-}$$

At anode:

$$PbO_2 + SO_4^{2-} + 4H^- + 2e^- \Rightarrow PbSO_4 + 2H_2O$$

Combining above two equation, the overall chemical reaction will be

$$PbO_2 + Pb + 2H_2SO_4 \Rightarrow 2PbSO_4 + 2H_2O$$

Charging Reactions:

When the battery is connected with the charger, the sulfuric acid molecules break into two ions, positive ions 2H+ and negative ions SO4-. The hydrogen exchange electrons with the cathode and become hydrogen, this hydrogen reacts with the PbSO4 in cathode and form Sulfuric Acid (H2SO4) and Lead (Pb). On the other hand, SO4- exchange electrons with anode and become radical SO4. This SO4 reacts with PbSO4 of anode and create the lead peroxide PbO2 and sulfuric acid (H2SO4). The energy gets stored by increasing the gravity of sulfuric acid and increasing the cell potential voltage.

As explained above, following chemical reactions takes place at Anode and Cathode during the charging process.

At cathode

$$PbSO_4 + 2e^- \Rightarrow Pb + SO_4^{2-}$$

At anode

$$PbSO_4 + 2H_2O \Rightarrow PbO_2 + SO_4^{2-} + 4H^- + 2e^-$$

Combining above two equation, the overall chemical reaction will be

$$2PbSO_4 + 2H_2O \Rightarrow PbO_2 + Pb + 2H_2SO_4$$

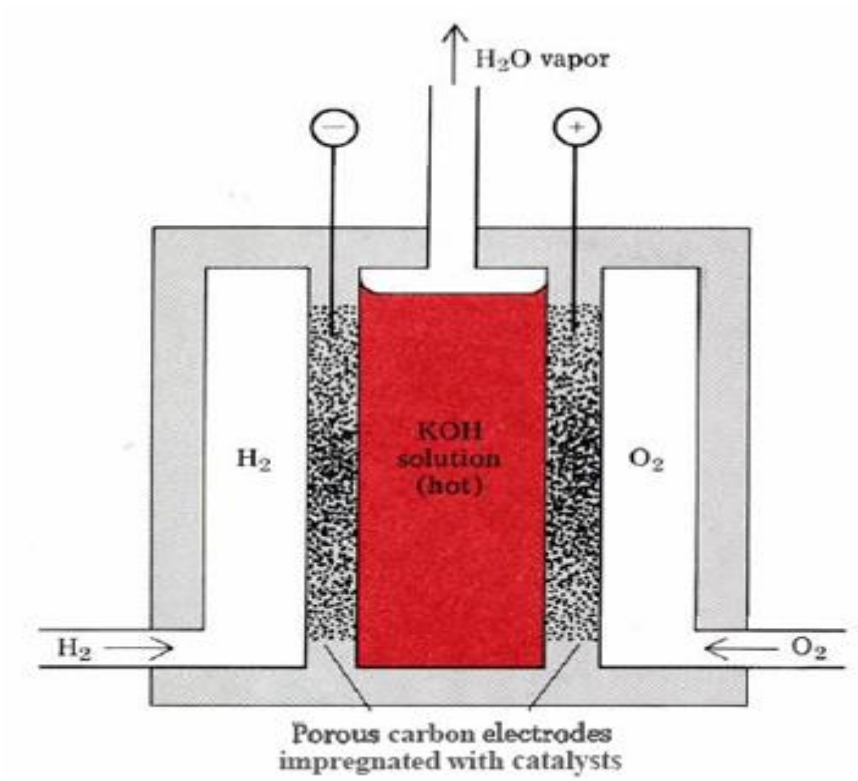
Lead-acid battery maintenance rules

1. Watering is the most neglected maintenance feature of flooded lead-acid batteries. As overcharging decreases water, we need to check it frequently. Less water creates oxidation in plates and decreases the lifespan of the battery. Add distilled or ionized water when needed.

2. Check for the vents, they need to be perfected with rubber caps, often the rubber caps sticks with the holes too tightly.
3. Recharge lead-acid batteries after each use. A long period without recharging provides sulfating in the plates.
4. Do not freeze the battery or charge it more than 49-degree centigrade. In cold ambient batteries need to be fully charged as fully charge batteries safer than the empty batteries in respect of freezing.
5. Do not deep discharge the battery less than 1.7V per cell.
6. To store a lead acid battery, it needs to be completely charged then the electrolyte needs to be drained. Then the battery will become dry and can be stored for a long time period.

FUEL CELL:

Hydrogen + Oxygen = Electricity + Water Vapor



Cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
Anode: $2H_2 \rightarrow 4H^+ + 4e^-$
Overall: $2H_2 + O_2 \rightarrow 2H_2O$

A fuel cell is a device that converts chemical potential energy (energy stored in molecular bonds) into electrical energy. A PEM (Proton Exchange Membrane) cell uses hydrogen gas (H_2) and oxygen gas (O_2) as fuel. The products of the reaction in the cell are water, electricity,

and heat. This is a big improvement over internal combustion engines, coal burning power plants, and nuclear power plants, all of which produce harmful by-products.

Since O_2 is readily available in the atmosphere, we only need to supply the fuel cell with H_2 which can come from an electrolysis process (see Alkaline electrolysis or PEM electrolysis).

There are four basic elements of a PEM Fuel Cell:

The anode, the negative post of the fuel cell, has several jobs. It conducts the electrons that are freed from the hydrogen molecules so that they can be used in an external circuit. It has channels etched into it that disperse the hydrogen gas equally over the surface of the catalyst.

The cathode, the positive post of the fuel cell, has channels etched into it that distribute the oxygen to the surface of the catalyst. It also conducts the electrons back from the external circuit to the catalyst, where they can recombine with the hydrogen ions and oxygen to form water.

As the name implies, the heart of the cell is the proton exchange membrane. It allows protons to pass through it virtually unimpeded, while electrons are blocked. So, when the H_2 hits the catalyst and splits into protons and electrons (remember, a proton is the same as an H+ ion) the protons go directly through to the cathode side, while the electrons are forced to travel through an external circuit. Along the way they perform useful work, like lighting a bulb or driving a motor, before combining with the protons and O_2 on the other side to produce water.

WORKING:

Pressurized hydrogen gas (H_2) entering the fuel cell on the anode side. This gas is forced through the catalyst by the pressure. When an H_2 molecule comes in contact with the platinum on the catalyst, it splits into two H+ ions and two electrons (e-). The electrons are conducted through the anode, where they make their way through the external circuit (doing useful work such as turning a motor) and return to the cathode side of the fuel cell.

Meanwhile, on the cathode side of the fuel cell, oxygen gas (O_2) is being forced through the catalyst, where it forms two oxygen atoms. Each of these atoms has a strong negative charge. This negative charge attracts the two H+ ions through the membrane, where they combine with an oxygen atom and two of the electrons from the external circuit to form a water molecule (H_2O).

Due to the high energetic content of hydrogen and high efficiency of fuel cells (55%), this great technology can be used in many applications like transport (cars, buses, forklifts, etc) and backup power to produce electricity during a failure of the electricity grid.

Advantages of the technology:

- By converting chemical potential energy directly into electrical energy, fuel cells avoid the "thermal bottleneck" (a consequence of the 2nd law of thermodynamics) and are thus inherently more efficient than combus

- By converting chemical potential energy directly into electrical energy, fuel cells avoid the "thermal bottleneck" (a consequence of the 2nd law of thermodynamics) and are thus inherently more efficient than combustion engines, which must first convert chemical potential energy into heat, and then mechanical work.
- Direct emissions from a fuel cell vehicle are just water and a little heat. This is a huge improvement over the internal combustion engine's litany of greenhouse gases.
- Fuel cells have no moving parts. They are thus much more reliable than traditional engines.
- Hydrogen can be produced in an environmentally friendly manner, while oil extraction and refining is very damaging.

☆ alkaline electrolyte

[O]: $H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$ (anode)

[R]: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ (cathode)

$$E_{cell}^{\theta} = E_R - E_O$$
$$= +0.40 - (-0.83)$$
$$= +1.23\ V$$

Module – 5: NANOMATERIALS AND NANOCHEMISTRY

It deals with the chemical application of nano materials along with their synthesis and characterization. Nano materials are mostly in the range of 10^{-9} m . Ex) Head of a pin--- 10^{6} nm.

Effect of size on material properties :

Property changes at nano scale wrt phisio-chemical, optical ,electrical and magnetic behavior.

1) Au nps change their colour by changing their size. 5 nm- red, 60nm -pink, 90 nm-violet.
2) A drop of water (5 nm) boils at 95.9^{0}c compared to normal water.
3) Adhesive property increases by decreasing the size as surface area increases.
4) Anti bacterial property increases as size decreases.
5) Optical properties of nps are dependent on size.

All these changes in nps compared to the bulk materials are due to a large s/v ratio which is given by,

$$S/V = 4\Pi r^2 / 4/3\Pi r^3 = 3/r$$

Materials with high s/v value possess very small diameter, porous and react with a much faster rate. Extra chemical reactivity is due to this high S/ V ratio.

Synthesis of nano materials : Two common methods followed are 1) Top- down and 2) Bottom –up approach.

Top-down Approach : It is a solid-state process involving size reduction from bulk materials. It mostly follows physical processes like crushing and grinding, milling.

Limitations : A slow process but not suitable for large scale applications/ leads to various defects and imperfections on the surface/ particles of uniform size and morphology not formed/

Some well known processes are electro-explosion, scattering, laser absorption

Bottom-up Approach.: It is purely a build-up one in which atom-by atom, molecule-molecule or cluster-cluster combination is made. Some of these are : sol-gel, chemical reduction, combustion, chemical vapor deposition, microwave synthesis etc. It also includes Green route which takes the help of sources like plants, yeast, fungi, bacteria etc.

Advantages: Generates particles of uniform size, shape, morphology./ particles of excellent quality and quantity/ an environmental friendly process

Advantages of plant extract:

Produces much less toxic waste/ act as both reducing and capping agent/ a safe process/ nps obtained have excellent medicinal properties.

Preparation of NPS :

The three raw materials needed are given below.

1) Precursor salt ; $AgNO_3$, $AuCl_3$, $ZnSO_4$, $CuSO_4$ etc.
2) Reducing Agent : citric acid, H_2O_2, Hydrazine. These are mostly phyto chemicals, plant metabolic, polyphenols.
3) Stabilizing agents : Also known as capping agents these are mostly polymers like PVA, PVP, surfactants.

The reducing agent provides electrons where as the capping agent makes the mixing process complete and faster.

Synthesis and characterization of gold nps :

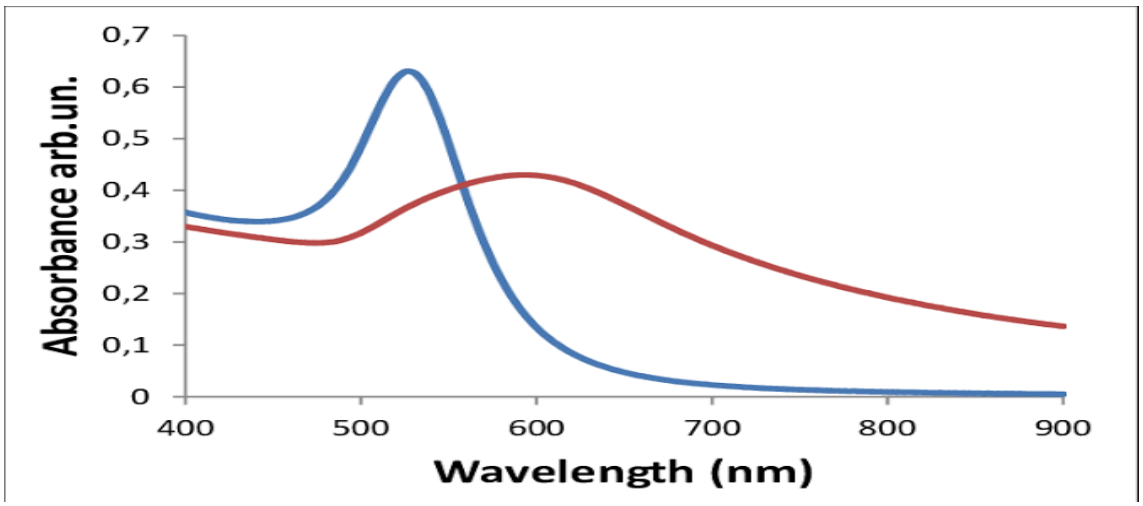
IMP) It is an environment friendly, low-cost and effective process. The phyto chemicals (Alkaloids, flavonoids) present in plant extract can act both as capping and reducing agent.

Synthesis: It is a bio reduction process in which leaves of Papaya and Tulasi are cut into small pieces. 10 gm of such leaves taken in a beaker with 200ml of distilled water was boiled for 30 minutes. Then it was cooled and filtered through 0.45mm membrane filter. Then the precursor salt is prepared by taking 20 mg of Gold(iii) chloride in 5 ml of distilled water. For the preparation of gold nps, 20 ml of plant extract was taken in a beaker and placed in a hot magnetic stirrer at 400c. Then 1 ml of gold salt was added to it drop wise. The solution turned pink indicating gold nps.

$$Au^{3+} + 3e^- \rightarrow Au^0$$

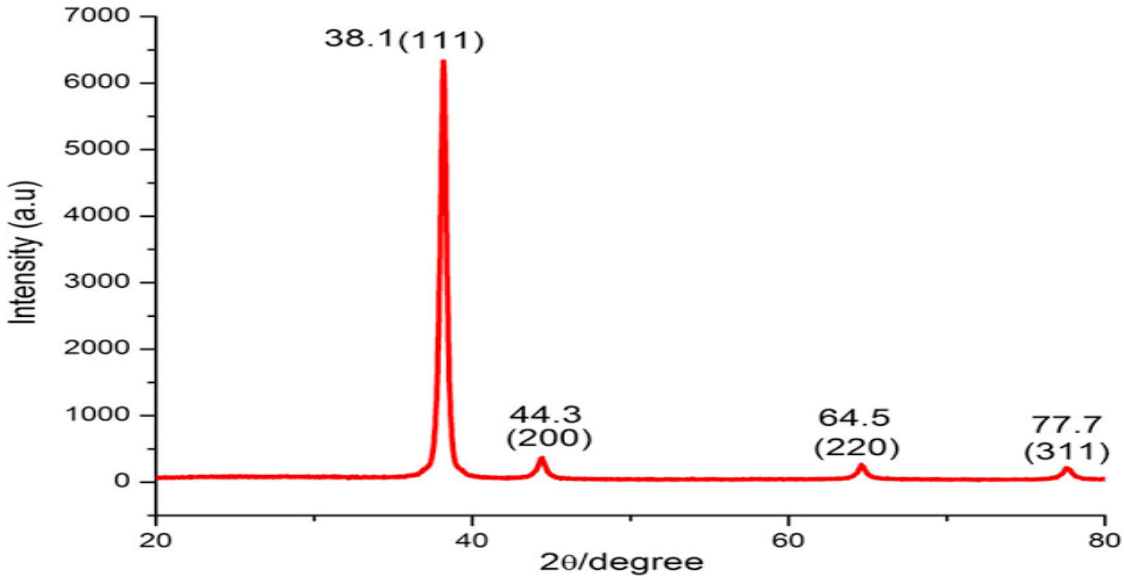
Characterisation:

UV-Visible Analysis : The absorption spectrum shows formation of Au as it shows a surface Plasmon resonance (SPR) band at550nm. Plasmons are sub particles arising out of quantization of plasma oscillations. SPR results from the resonant oscillation of free electrons at the interface between positive and negative permittivity mutually stimulated by light. It is otherwise a manifestation of a coherent oscillation of conduction band electrons.



X-ray diffraction study : XRD pattern shows the production of Au nps by the reduction of Au3+ to Au . It identifies the crystallographic nature of nps. The peak (111) at 2ɵ= 38° shows nps with fcc structure. Calculating the crystalline size (d) using Debye-Scheres equation,

$d = 0.94\lambda / \beta\cos\theta \approx 15nm = K\lambda / \beta\cos\theta$,K≈ 1, ɵ = Bragg angle or diffraction angle



Synthesis, characterization and application of Ag nps :

1) Preparation of plant extract : 20 gm of finely divided NEEM leaves were boiled with100 ml of double distilled water for 30 minutes. It was then filtered using a 0.45mm membrane filter to get the extract.

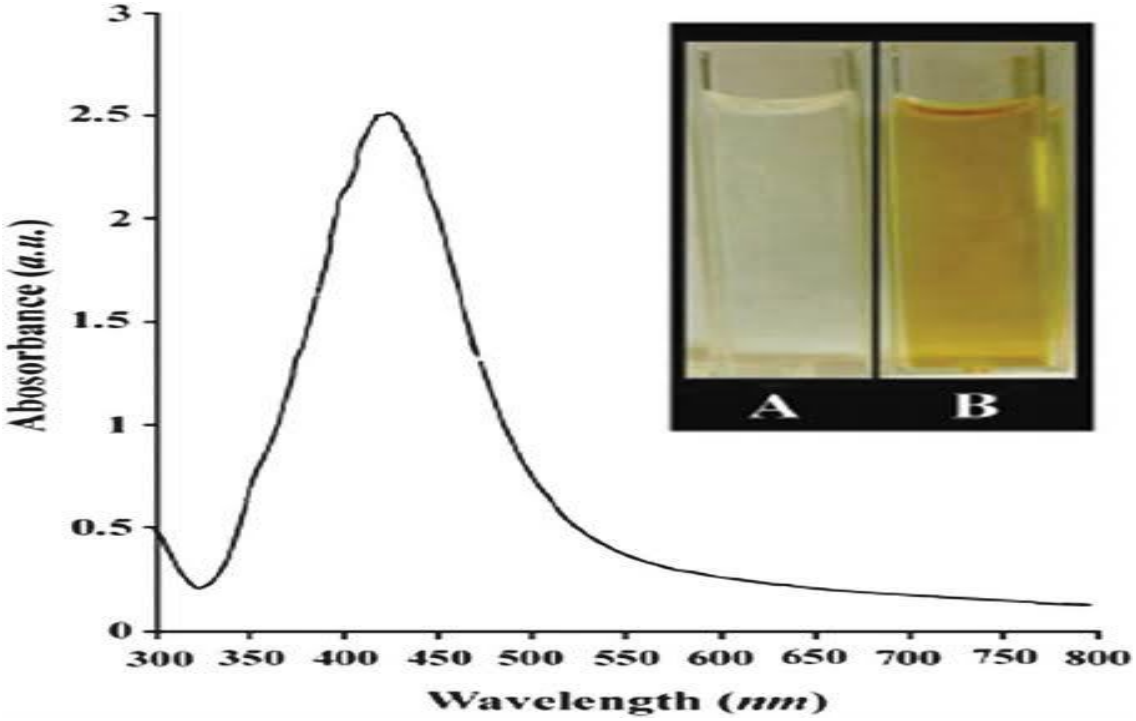
2) Preparation of AgNO3 solution: Suitable amount of $AgNO_3$ salt was added to double distilled water to prepare 10 ml of the solution after boiling.

3)Synthesis of Ag np ; 5 ml of extract was taken inside a 50 ml beaker and placed in a magnetic stirrer. Then 1 ml of above salt solution was added to it drop wise using a micropipette. The colour suddenly changed to yellowish brown indicating formation of nps.

$Ag^+ + e^- \longrightarrow Ag$

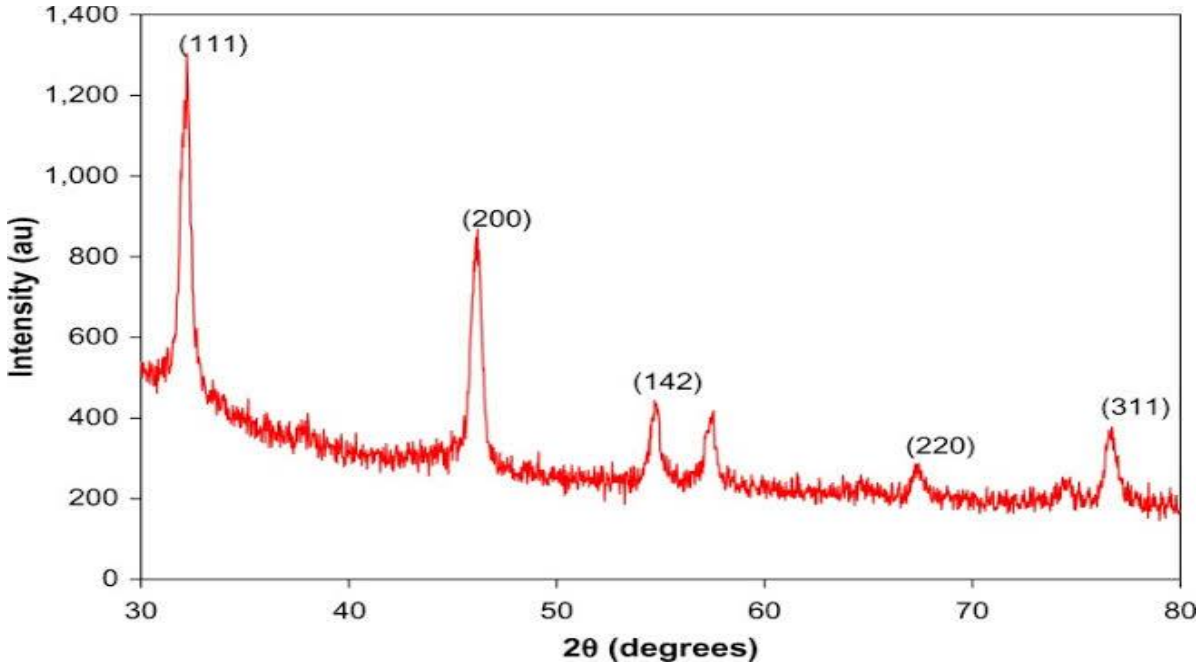
Characterisation : The sample was characterized using uv-vis spectroscopy and x-ray diffraction study.

UV-VIS study : The absorption maxima found at 400nm indicates formation of $AgNO_3$ nps.



XRD STUDY: It provides information about the crystalline nature and size of nps. The peak at 2ɵ =38^0 corresponds to(111) plane of nps and reveals fcc structure of silver nps. The size of nps can be calculated by Debye-Scherer equation,

$$d = 0.94\lambda/\beta \cos \theta \approx 13nm \quad (\beta = 0.001 - 0.02)$$



Application: Food packaging, cosmetic industry, bio-sensors, catalysis, coating materials, textile industry.

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